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Computation and analysis of surface wave dispersion and attenuation in layered viscoelastic-vertical transversely isotropic media by the generalized R/T coefficient method

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SUMMARY

In this study, we propose a systematic and effective method, that is, an extended version of the generalized reflection/transmission (R/T) coefficient method, for computing the phase-velocity (c_r) dispersion curves, attenuation coefficient (α) curves, and eigenfunctions of both Rayleigh and Love waves as well as the ellipticity of Rayleigh waves in layered viscoelastic-vertical transversely isotropic (VTI) media. The numerical scheme of combining the root-searching method with the local optimization method is designed for determining the complex-valued modal solutions (i.e. complex wavenumber $k = \omega/c_r - i\alpha$) of surface waves. The near-surface sedimentary geological environment is taken as the model example because it is typical viscoelastic-VTI media. Besides the anisotropic-viscoelastic (AV) media, our algorithm can also compute surface waves in isotropic-elastic (IE), isotropic-viscoelastic (IV) and anisotropic-elastic (AE) media by resetting the corresponding parameters. Using the six-layer half-space models and in these four media, we verify the correctness of our algorithm by benchmarking the modal solutions against those from other methods. In the four-layer half-space model, by comparing the results of IE, IV, AE and AV media, we analyse the effects of velocity anisotropy, viscoelasticity and attenuation anisotropy on the dispersion and attenuation characteristics of both Rayleigh and Love waves in detail. Our study can provide a theoretical basis and useful tool for surface wave imaging considering the anisotropy and/or viscoelasticity of the medium, which has the potential to better investigate the solid Earth's internal structure.

Key words: Elasticity and anelasticity; Seismic anisotropy; Seismic attenuation; Surface waves and free oscillations.

1 INTRODUCTION

Surface wave imaging, such as Rayleigh or Love wave imaging, is one of the important geophysical imaging methods and has been widely used for investigating the internal structure of solid Earth with the near-surface, regional and global scales (e.g. Shapiro *et al.* 2005; Socco *et al.* 2010; Li *et al.* 2019; Wu *et al.* 2020). It is based on two important characteristics of surface waves, namely, velocity dispersion and amplitude attenuation. The classic surface-wave dispersion imaging is under the assumption of isotropic-elastic (IE) media and mainly estimates the subsurface shear-wave velocity (*Vs*) structure by extracting and inverting the phase-velocity dispersion curves of surface waves (e.g. Xia *et al.* 1999; Yao *et al.* 2006; Yang *et al.* 2007). However, the solid Earth, especially its near-surface sediments, has been extensively proven to be anisotropic and viscoelastic, which implies large effects on the dispersion and attenuation of surface waves (Carcione 2015; Yuan *et al.* 2018; Borcherdt 2020, 2022).

On the one hand, taking the velocity anisotropy of the medium into account, surface wave dispersion imaging is further extended to detect the radial and azimuthal anisotropic structures of *Vs* (e.g. Yao *et al.* 2010; Xie & Liu 2015; Hu *et al.* 2020). In general, the radial anisotropy (or polarization anisotropy) of *Vs* [i.e. the discrepancies between the horizontally (*Vsh*) and vertically (*Vsv*) polarized *Vs*], which accords with the assumption of the transversely isotropic (TI) medium with a vertical symmetry axis (the so-called VTI medium), can be determined by inverting the phase-velocity dispersion curves of Rayleigh and Love waves separately or jointly

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(e.g. Shapiro *et al.* 2004; Xie *et al.* 2013; Hu *et al.* 2020). Based on the assumption of the TI medium with a horizontal symmetry axis (the so-called HTI medium), the perturbation theory derived from Rayleigh's principle (Smith & Dahlen 1973; Montagner & Nataf 1986), which builds the connection between the backazimuth-dependent Rayleigh wave phase velocity dispersion and the depth-dependent azimuthal anisotropic parameters, is currently widely used in the inversion of the azimuthal anisotropy of *Vs* (e.g. Lin *et al.* 2010; Yao *et al.* 2010). In addition, shear wave splitting is also a good tool for the anisotropy study of *Vs* (e.g. Diaz-Acosta *et al.* 2023). On the other hand, the viscoelasticity of the medium will weaken the amplitude of propagating surface waves through energy dissipation, which is usually known as intrinsic attenuation (Knopoff 1964; Aki & Richards 1980). Surface wave attenuation imaging is developed to retrieve the subsurface shear-wave quality factor (Qs) structure by extracting and inverting the attenuation coefficient curves of surface waves (e.g. Xia *et al.* 2002; Yang & Forsyth 2008; Liu *et al.* 2022). Some trials for the shallow structure have also shown that joint imaging of surface wave dispersion and attenuation can simultaneously and more accurately reveal *Vs* and *Qs* structures (e.g. Lai *et al.* 2002b; Misbah & Strobbia 2014; Armstrong *et al.* 2020; Bergamo *et al.* 2023).

Computation of surface wave modal solutions (i.e. wavenumber k; $k = \omega/c_r$ for elastic medium, while $k = \omega/c_r - i\alpha$ for viscoelastic medium), that is forward modelling of phase-velocity (c_r) dispersion curves and attenuation coefficient (α) curves of surface waves, is the theoretical basis of surface wave imaging. For IE layered media, besides the numerical discretization method (e.g. Shi *et al.* 2022), it is a more widely used method that the modal solutions of surface waves can be robustly and efficiently determined by using a suitable root-searching scheme (e.g. bisection method) to compute the roots of the dispersion equation or the zero points of the secular function, currently. Deriving the dispersion equation (or secular function) from the source-free dynamic equation mainly involves two types of widely used methods. The one is the propagator-matrix-based methods, including the classic Thomson–Haskell method (Thomson 1950; Haskell 1953) and its various improved versions (e.g. Knopoff 1964; Dunkin 1965; Watson 1970; Panza 1985; Buchen & Ben-Hador 1996). Another is the reflection/transmission (R/T)-coefficient-based methods, which were first proposed by Kennett & Kerry (1979). Adopting the R/T coefficients of Luco & Apsel (1983), Chen (1993) proposed a more concise and self-contained version, namely the generalized R/T coefficient method (GRTM), for calculating both the dispersion curves and the eigenfunctions. Introducing a family of secular functions can further improve the GRTM for solving the numerical difficulty of finding the trapped modes in the model having low-velocity layers (He *et al.* 2006; Wu & Chen 2016).

Compared with the case of IE media, the stiffness matrix in anisotropic-elastic (AE) media contains more independent components. To calculate the surface wave modal solutions, the formula of the secular function needs to be deduced again. Extending the Thomson–Haskell method, Anderson (1961) proposed the method of calculating the modal solutions of surface waves in VTI layered media. Crampin (1970) extended the matrix procedure developed by the Thomson–Haskell method to obtain the phase velocity dispersion relation for generalized surface waves in multilayered media, where the layers may be either isotropic or anisotropic. Ke *et al.* (2011) and Ikeda & Matsuoka (2013) put forward the improved versions of Anderson (1961) by using the modified Thomson–Haskell method and the reduced delta matrix method, respectively. Extending the GRTM of Chen (1993), Park (1996) calculated the surface waves in layered anisotropic media with an arbitrarily oriented axis of symmetry, while Ji *et al.* (2020) simulated the dispersion curves of Rayleigh channel waves in the coal seam model with elastic-TI media, and analysed the effects of Thomsen's parameters (Thomsen 1986). However, their derivation process for the GRTM is not entirely explicit and there is a lack of algorithm validation.

In addition, unlike the case of IE media, in isotropic-viscoelastic (IV) media, the components of the stiffness matrix are complex-valued and frequency-dependent. Consequently, the modal solutions and secular function of surface waves also become complex-valued. On this condition, the key problem is how to exactly determine the zero points of a complex-valued function of a complex variable. Based on the Cauchy residue theorem of complex analysis, Lai & Rix (2002a) proposed an effective method to compute the dispersion and attenuation curves as well as eigenfunctions of Rayleigh waves. Taking the kernel function (Chen 1999) as a new secular function, Pan *et al.* (2022) proposed the modified GRTM, which can sequentially determine phase velocity and attenuation coefficient of surface waves in the real domain. Moreover, although requiring the given initial values, iteration methods are the simplest to solve this problem, for example both Muller's method and Newton's method have been successfully utilized to determine the complex-valued modal solutions of surface waves (Cao & Dong 2010; Zhang *et al.* 2016). In summary, there are no relevant reports on the computation method of surface-wave modal solutions in a more realistic case, that is, in anisotropic-viscoelastic (AV) media, at present.

The purpose of this work is to propose a systematic and effective method, as an extension version of the GRTM (Chen 1993; Wu & Chen 2016), to compute the modal solutions and eigenfunctions of both Rayleigh and Love waves as well as the ellipticity of Rayleigh waves in layered viscoelastic-VTI media, and investigate the dispersion and attenuation characteristics of surface waves. The properties of both velocity and attenuation anisotropy of the medium are described by the VTI medium model (e.g. Thomsen 1986; Zhu & Tsvankin 2006), which is widely used in the study of radial anisotropy based on surface waves and is the simplest and the most common anisotropy type for subsurface structure (e.g. sedimentary strata). The viscoelasticity of the medium can be characterized by different viscoelastic rheological models, including constant-Q models and nearly constant-Q models (e.g. Carcione 2015). In this study, we adopt Kjartansson's constant-Q model (Kjartansson 1979), which is popular and widely used in the forward and inversion research of seismic waves, because it enables us to perfectly describe the frequency-independent Q anelastic behaviour using less parameters (e.g. Zhu & Carcione 2014).

The article is organized as follows. We first introduce the methodology in Section 2, which involves the constitutive relationship in viscoelastic-VTI media, as well as the concepts of generalized R/T coefficients and a family of secular functions. Both the *P*-SV and SH wave cases are considered, and the explicit expressions are given in the entire derivation process. Then, in Section 3, we list the steps of algorithm implementation, in which the complex-valued modal solutions are determined by combining the root-searching method with the local optimization method. By comparing the modal solutions from our GRTM with those from other methods, we also conduct algorithm validation in IE, IV, AE and AV media, respectively. In Section 4, we explore the effects of velocity anisotropy, viscoelasticity and attenuation anisotropy on the dispersion and attenuation characteristics of surface waves in detail by comparing the results of IE, IV, AE and AV media. Finally, we make some conclusions.

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2 METHODOLOGY

The modal solutions of surface waves are non-trivial solutions of the source-free dynamic equation (without source term) under the boundary conditions of the free surface, continuity at each interface and radiation conditions (Aki & Richards 1980). This section will introduce the methodology involved in calculating the surface-wave modal solutions. The cylindrical coordinate system is adopted in this study, and the $r-o-\theta$ plane is horizontally placed while the z-direction is vertically downward. As Fig. 1 shows, we are concerned with the horizontally homogeneous layered media with N layers over a half-space. The jth layer is sandwiched between two horizontal flat interfaces, that is, $z = z^{(j-1)}$ and $z = z^{(j)}$, $j = 1, 2, \ldots, N, N+1$; the top layer has a free surface, that is, $z = z^{(0)} = 0$; for the half-space layer, the bottom interface can be considered as $z = z^{(N+1)} = +\infty$.

2.1 Constitutive relationship in viscoelastic-VTI media

The most generalized constitutive relationship between strain and stress for anisotropic and linear viscoelastic media in the frequency domain can be expressed as (Carcione 1995):

$$\mathbf{T}^{(j)}(\mathbf{x},\omega) = \mathbf{M}^{(j)}(\mathbf{x},\omega)\mathbf{S}^{(j)}(\mathbf{x},\omega),$$
$$\left(T_m^{(j)}(\mathbf{x},\omega) = M_{mn}^{(j)}(\mathbf{x},\omega)S_n^{(j)}(\mathbf{x},\omega),\ m,n = 1,2,\ldots,6\right), \tag{1}$$

where $\mathbf{T}^{(j)} = [\sigma_{rr}^{(j)}, \sigma_{\theta\theta}^{(j)}, \sigma_{zz}^{(j)}, \sigma_{\theta z}^{(j)}, \sigma_{rz}^{(j)}, \sigma_{r\theta}^{(j)}]^{\mathrm{T}}$ and $\mathbf{S}^{(j)} = [\varepsilon_{rr}^{(j)}, \varepsilon_{\theta\theta}^{(j)}, \varepsilon_{zz}^{(j)}, 2\varepsilon_{\theta z}^{(j)}, 2\varepsilon_{rz}^{(j)}, 2\varepsilon_{r\theta}^{(j)}]^{\mathrm{T}}$ are the stress vector and strain vector, respectively; the superscript T denotes the transpose; $\mathbf{x} = (r, \theta, z)$ is the 3-D spatial position vector; ω is angular frequency; $\mathbf{M}^{(j)}$ is the symmetric complex-modulus stiffness matrix. For viscoelastic-VTI media, in abbreviated Voigt notation, $\mathbf{M}^{(j)}$ can be given in the following form by five independent complex moduli (i.e. $M_{11}^{(j)}$, $M_{13}^{(j)}$, $M_{33}^{(j)}$, $M_{55}^{(j)}$ and $M_{66}^{(j)}$, see Fig. 1):

$$\mathbf{M}^{(j)} = \begin{bmatrix} M_{11}^{(j)} & M_{11}^{(j)} - 2M_{66}^{(j)} & M_{13}^{(j)} & 0 & 0 & 0 \\ M_{11}^{(j)} - 2M_{66}^{(j)} & M_{11}^{(j)} & M_{13}^{(j)} & 0 & 0 & 0 \\ M_{13}^{(j)} & M_{13}^{(j)} & M_{33}^{(j)} & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{55}^{(j)} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{55}^{(j)} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{66}^{(j)} \end{bmatrix}. \tag{2}$$

In this study, the viscoelasticity of the medium is described by Kjartansson's constant-Q model (Kjartansson 1979), which is one of the widely used viscoelastic rheological models. Analogous to the isotropic cases, the frequency-dependent complex modulus ($M_{mn}^{(j)}(\omega)$) of this model can be generalized for anisotropic media as follows:

$$M_{mn}^{(j)}(\omega) = C_{mn}^{(j)}\cos^2\left(\pi\upsilon_{mn}^{(j)}/2\right)\left(\frac{i\omega}{\omega_r}\right)^{2\upsilon_{mn}^{(j)}},$$
$$\upsilon_{mn}^{(j)} = \frac{\arctan\left(1/Q_{mn}^{(j)}\right)}{\pi}, \tag{3}$$

where $\omega_r = 2\pi f_r$ is reference angular frequency, and f_r is reference frequency; i denotes imaginary unit; $C_{mn}^{(j)}$ is elasticity stiffness component (or elasticity constant); $Q_{mn}^{(j)}$ is quality-factor (Q) component; $\upsilon_{mn}^{(j)}$ is a dimensionless parameter related to $Q_{mn}^{(j)}$, and $0 < \upsilon_{mn}^{(j)} < 1/2$ for a positive Q value. Corresponding to the complex modulus, the complex velocity, phase velocity and quality factor can be defined by:

$$V_{mn}^{(j)}(\omega) = \sqrt{\frac{M_{mn}^{(j)}(\omega)}{\rho^{(j)}}}, \tag{4}$$
$$V_{p,mn}^{(j)}(\omega) = 1/\mathrm{Re}\left[1/V_{mn}^{(j)}(\omega)\right], \tag{5}$$
$$Q_{mn}^{(j)} = \frac{\mathrm{Re}\left[M_{mn}^{(j)}(\omega)\right]}{\mathrm{Im}\left[M_{mn}^{(j)}(\omega)\right]} = \frac{1}{\tan(\pi\upsilon_{mn}^{(j)})}, \tag{6}$$

where $\rho^{(j)}$ is density; the operators Re[·] and Im[·] mean taking the real and imaginary parts of a complex variable, respectively; the subscript letter p represents the English word phase. Eq. (6) confirms that $Q_{mn}^{(j)}$ is frequency-independent (i.e. constant-Q).

In viscoelastic-VTI media, the anisotropy of the media can not only exhibit velocity anisotropy but also exhibit attenuation anisotropy. Anisotropic velocity can be described by the elasticity stiffness matrix. Corresponding to the complex modulus (eqs 2 and 3), five independent elasticity constants (i.e. $C_{11}^{(j)}$, $C_{13}^{(j)}$, $C_{33}^{(j)}$, $C_{55}^{(j)}$ and $C_{66}^{(j)}$) compose the elasticity stiffness matrix, and they can be expressed as (Thomsen 1986):

$$C_{33}^{(j)} = \rho^{(j)}\left(V_{P0}^{(j)}\right)^2, C_{55}^{(j)} = \rho^{(j)}\left(V_{S0}^{(j)}\right)^2, \tag{7a}$$
$$C_{11}^{(j)} = \left(1+2\varepsilon^{(j)}\right)C_{33}^{(j)},\ C_{66}^{(j)} = \left(1+2\gamma^{(j)}\right)C_{55}^{(j)}, \tag{7b}$$
$$C_{13}^{(j)} = C_{33}^{(j)}\sqrt{\left(1-\frac{C_{55}^{(j)}}{C_{33}^{(j)}}\right)\left(1-\frac{C_{55}^{(j)}}{C_{33}^{(j)}}+2\delta^{(j)}\right)} - C_{55}^{(j)}, \tag{7c}$$

where $V_{P0}^{(j)}$ and $V_{S0}^{(j)}$ are the P-wave and S-wave velocities along the symmetry axis of TI media, respectively; $\varepsilon^{(j)}$, $\delta^{(j)}$ and $\gamma^{(j)}$ are three dimensionless Thomsen's parameters that describe the velocity anisotropy of TI media (Thomsen 1986); $\varepsilon^{(j)}$ and $\gamma^{(j)}$ control the velocity anisotropy degree of P and S waves, respectively; $\delta^{(j)}$ reflects the velocity difference of P and S waves.

Anisotropic attenuation can be described by the quality factor matrix, and it has the same matrix form as the elastic stiffness matrix (Zhu & Tsvankin 2006). For the VTI media with VTI attenuation, the quality factor matrix consists of five independent components (i.e. $Q_{11}^{(j)}$, $Q_{13}^{(j)}$, $Q_{33}^{(j)}$, $Q_{55}^{(j)}$ and $Q_{66}^{(j)}$), as follows:

$$Q_{33}^{(j)} = Q_{P0}^{(j)},\ Q_{55}^{(j)} = Q_{S0}^{(j)}, \tag{8a}$$
$$Q_{11}^{(j)} = \frac{1}{1+\varepsilon_Q^{(j)}}Q_{33}^{(j)},\ Q_{66}^{(j)} = \frac{1}{1+\gamma_Q^{(j)}}Q_{55}^{(j)}, \tag{8b}$$
$$Q_{13}^{(j)} = \frac{2Q_{33}^{(j)}C_{13}^{(j)}\left(C_{13}^{(j)}+C_{55}^{(j)}\right)}{\delta_Q^{(j)}C_{33}^{(j)}\left(C_{33}^{(j)}-C_{55}^{(j)}\right) - \frac{\left(Q_{33}^{(j)}-Q_{55}^{(j)}\right)C_{55}^{(j)}\left(C_{13}^{(j)}+C_{33}^{(j)}\right)^2}{Q_{55}^{(j)}\left(C_{33}^{(j)}-C_{55}^{(j)}\right)} + 2C_{13}^{(j)}\left(C_{13}^{(j)}+C_{55}^{(j)}\right)}, \tag{8c}$$

where $Q_{P0}^{(j)}$ and $Q_{S0}^{(j)}$ are the P-wave and S-wave quality factors along the symmetry axis of TI media, respectively; $\varepsilon_Q^{(j)}$, $\delta_Q^{(j)}$ and $\gamma_Q^{(j)}$ are three dimensionless Thomsen-style parameters defined by Zhu & Tsvankin (2006) for the description of the attenuation anisotropy in viscoelastic TI media; $\varepsilon_Q^{(j)}$ and $\gamma_Q^{(j)}$ control the attenuation anisotropy degree of P and S waves, respectively; $\delta_Q^{(j)}$ reflects the attenuation difference of P and S waves.

The velocity and attenuation anisotropy of body waves can be reflected by the characteristic solutions (eqs A4–A6) of the Kelvin–Christoff equation (eq. A2) in viscoelastic-VTI medium (see Appendix A). Taking a homogeneous full-space medium as an example

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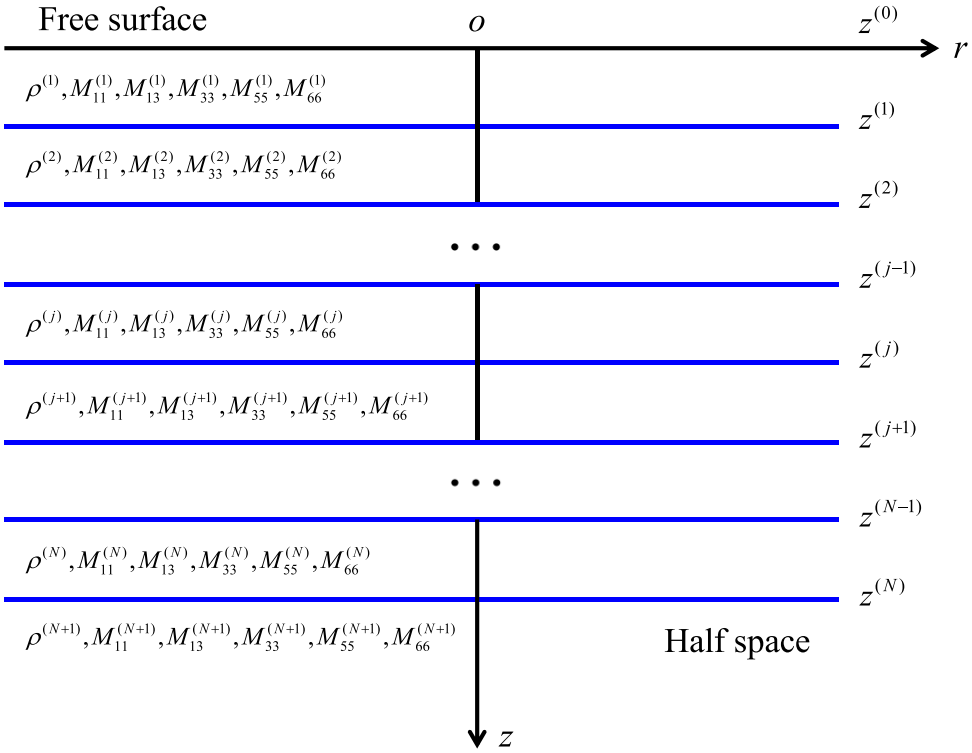


Figure 1. Configuration and coordinate system of the multilayered half-space model with viscoelastic-VTI media. $\rho^{(j)}$ is the density; $M_{11}^{(j)}$, $M_{13}^{(j)}$, $M_{33}^{(j)}$, $M_{55}^{(j)}$ and $M_{66}^{(j)}$ are five independent complex moduli of the jth layer (see eqs 2 and 3), $j = 1, 2, \ldots, N, N+1$.

($V_{S0} = 200$ m s^{-1}, $V_{P0} = 800$ m s^{-1}, $\rho = 2.0$ g cm^{-3}, $Q_{S0} = 20$, $Q_{P0} = 40$, $\varepsilon = 0.1$, $\delta = 0.05$, $\gamma = 0.1$, $\varepsilon_Q = 0.4$, $\delta_Q = 0.1$, $\gamma_Q = 0.4$ and $f_r = 25$ Hz), Fig. 2 shows the variations of quality factors, phase velocities and attenuation coefficients of quasi-P, quasi-SV and SH waves with propagation angle θ in the $x-z$ plane at the frequency of 25 Hz. It demonstrates that the velocity dispersion and attenuation properties of body waves are direction-dependent in the viscoelastic-VTI medium. Surface waves are derived from body waves, and these properties are also transmitted from body waves to surface waves.

2.2 Generalized R/T coefficients

As introduced in Appendix B, by combining the constitutive equation (eq. 1) with the momentum conservation equation (without source term) and geometric equation, we can derive the governing equations used in this study (i.e. eqs B3 and B6). Then, following standard theoretical derivation, we can obtain two systems of first-order ordinary differential equations corresponding to the two decoupled cases of P-SV and SH waves (i.e. eq. B7), as well as their general solution (i.e. eq. B12). To determine the unknown vector $\mathbf{C}^{(j)}$ in general solution, we shall introduce the concept of generalized reflection and transmission coefficients, and derive the corresponding formulae to calculate them (Chen 1993; Pei *et al.* 2008; Wu & Chen 2016).

For P-SV waves, the general solution (i.e. eq. B12) can be further expressed as:

$$\mathbf{y}^{(j)}(z) = \begin{pmatrix} \mathbf{E}_{11}^{(j)} & \mathbf{E}_{12}^{(j)} \\ \mathbf{E}_{21}^{(j)} & \mathbf{E}_{22}^{(j)} \end{pmatrix} \begin{pmatrix} \mathbf{\Lambda}_d^{(j)}(z) & \\ & \mathbf{\Lambda}_u^{(j)}(z) \end{pmatrix} \begin{pmatrix} \mathbf{C}_d^{(j)} \\ \mathbf{C}_u^{(j)} \end{pmatrix}, \tag{9}$$

where $\mathbf{E}_{11}^{(j)}$, $\mathbf{E}_{12}^{(j)}$, $\mathbf{E}_{21}^{(j)}$ and $\mathbf{E}_{22}^{(j)}$ are submatrices of matrix $\mathbf{E}^{(j)}$; $\mathbf{\Lambda}_d^{(j)}(z)$ and $\mathbf{\Lambda}_u^{(j)}(z)$, which are submatrices of matrix $\mathbf{\Lambda}^{(j)}$, represent the propagation factors of downgoing and upgoing waves, respectively; $\mathbf{C}_d^{(j)} = [\, C_{Pd}^{(j)} \; C_{SVd}^{(j)} \,]^T$ and $\mathbf{C}_u^{(j)} = [\, C_{Pu}^{(j)} \; C_{SVu}^{(j)} \,]^T$ are submatrices of matrix $\mathbf{C}^{(j)}$, and correspond to the P-SV wave amplitude vectors of the downgoing and upgoing waves, respectively. The explicit expressions of $\mathbf{E}^{(j)}$and $\mathbf{\Lambda}^{(j)}$ are given in Appendix C.

For upgoing waves, as Fig. 3(a) shows, the generalized reflection and transmission coefficients ($\hat{\mathbf{T}}_u^{(j)}$ and $\hat{\mathbf{R}}_{ud}^{(j)}$) at the layer interface $z = z^{(j)}$ are defined as:

$$\begin{cases} \mathbf{C}_u^{(j)} = \hat{\mathbf{T}}_u^{(j)} \mathbf{C}_u^{(j+1)} \\ \mathbf{C}_d^{(j+1)} = \hat{\mathbf{R}}_{ud}^{(j)} \mathbf{C}_u^{(j+1)} \end{cases}. \tag{10}$$

According to the continuity boundary conditions (i.e. $\mathbf{y}^{(j)}(z^{(j)}) = \mathbf{y}^{(j+1)}(z^{(j)})$), we have:

$$\begin{aligned} &\begin{pmatrix} \mathbf{E}_{11}^{(j)} & \mathbf{E}_{12}^{(j)} \\ \mathbf{E}_{21}^{(j)} & \mathbf{E}_{22}^{(j)} \end{pmatrix} \begin{pmatrix} \mathbf{\Lambda}_d^{(j)}(z^{(j)}) & \\ & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{C}_d^{(j)} \\ \mathbf{C}_u^{(j)} \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{E}_{11}^{(j+1)} & \mathbf{E}_{12}^{(j+1)} \\ \mathbf{E}_{21}^{(j+1)} & \mathbf{E}_{22}^{(j+1)} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \\ & \mathbf{\Lambda}_u^{(j+1)}(z^{(j)}) \end{pmatrix} \begin{pmatrix} \mathbf{C}_d^{(j+1)} \\ \mathbf{C}_u^{(j+1)} \end{pmatrix}. \end{aligned} \tag{11}$$

Submitting eq. (10) to eq. (11), we can derive:

$$\begin{aligned} &\begin{pmatrix} \mathbf{O}_{11}^{(j)} & \mathbf{O}_{12}^{(j)} \\ \mathbf{O}_{21}^{(j)} & \mathbf{O}_{22}^{(j)} \end{pmatrix} \begin{pmatrix} \mathbf{\Lambda}_d^{(j)}(z^{(j)}) \hat{\mathbf{R}}_{ud}^{(j-1)} \\ \mathbf{I} \end{pmatrix} \hat{\mathbf{T}}_u^{(j)} \mathbf{C}_u^{(j+1)} \\ &= \begin{pmatrix} \hat{\mathbf{R}}_{ud}^{(j)} \\ \mathbf{\Lambda}_u^{(j+1)}(z^{(j)}) \end{pmatrix} \mathbf{C}_u^{(j+1)}, \end{aligned} \tag{12}$$

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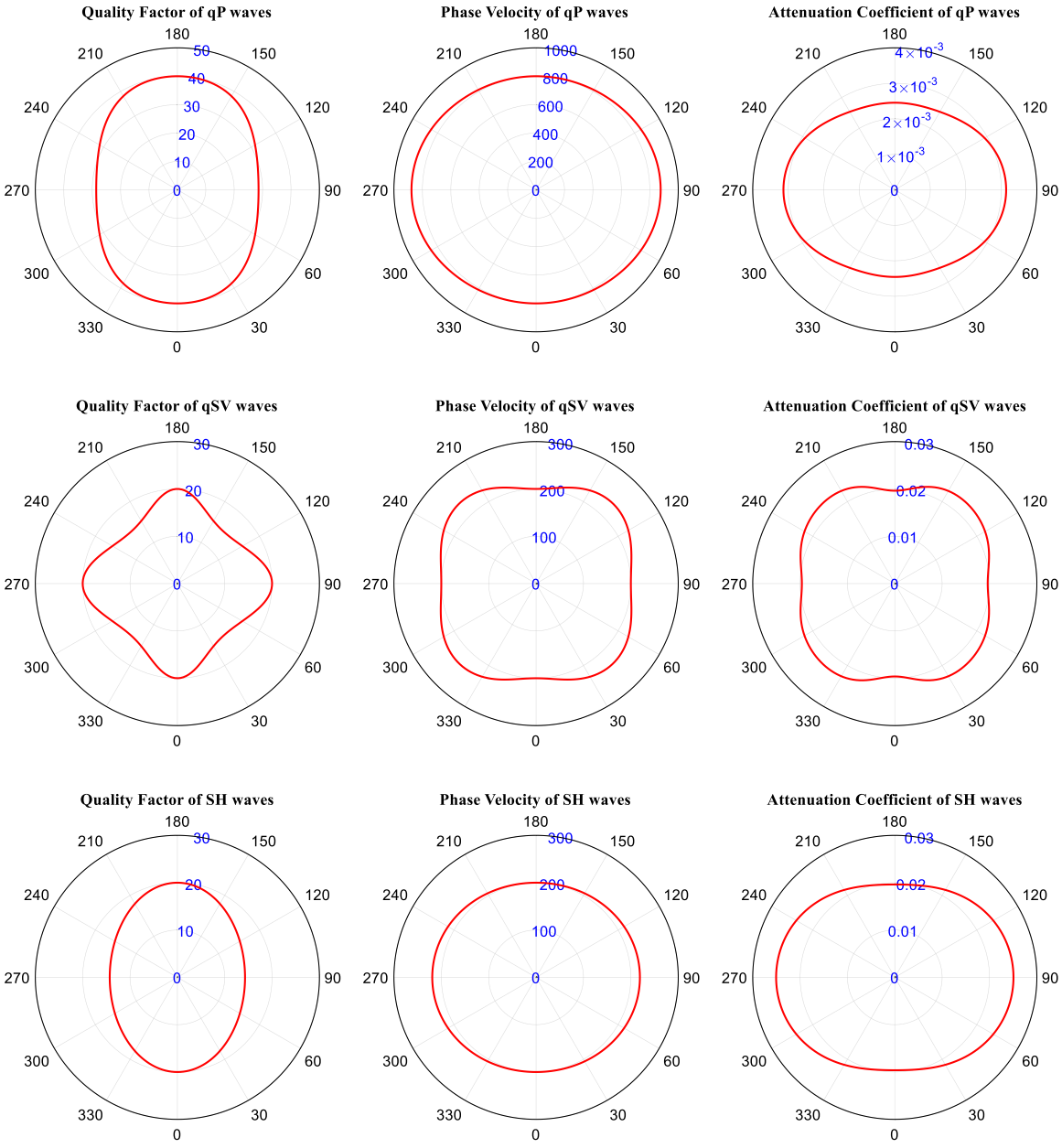


Figure 2. The variations of quality factors (left-hand panel), phase velocities (middle panel) and attenuation coefficients (right-hand panel) of quasi-*P*, quasi-SV and SH waves with propagation angle θ in the $x-z$ plane at the frequency of 25 Hz.

where $\mathbf{O}^{(j)} = (\mathbf{E}^{(j+1)})^{-1}\mathbf{E}^{(j)}$ is a known matrix, and $\mathbf{O}_{11}^{(j)}$, $\mathbf{O}_{12}^{(j)}$, $\mathbf{O}_{21}^{(j)}$ and $\mathbf{O}_{22}^{(j)}$ are its submatrices. Further organizing eq. (12), the formulae of $\hat{\mathbf{T}}_u^{(j)}$ and $\hat{\mathbf{R}}_{ud}^{(j)}$ can be obtained:

$$\begin{cases} \hat{\mathbf{T}}_u^{(j)} = \left[\mathbf{O}_{21}^{(j)}\mathbf{\Lambda}_d^{(j)}(z^{(j)})\hat{\mathbf{R}}_{ud}^{(j-1)} + \mathbf{O}_{22}^{(j)}\right]^{-1}\mathbf{\Lambda}_u^{(j+1)}(z^{(j)}) \\ \hat{\mathbf{R}}_{ud}^{(j)} = \left[\mathbf{O}_{11}^{(j)}\mathbf{\Lambda}_d^{(j)}(z^{(j)})\hat{\mathbf{R}}_{ud}^{(j-1)} + \mathbf{O}_{12}^{(j)}\right]\hat{\mathbf{T}}_u^{(j)} \end{cases}, \tag{13}$$

where $j = 1, 2, \ldots, N-2, N-1$. According to eq. (13), to calculate $\hat{\mathbf{T}}_u^{(j)}$ and $\hat{\mathbf{R}}_{ud}^{(j)}$ for the arbitrary layer interface, it is necessary to know the reflection coefficient $\hat{\mathbf{R}}_{ud}^{(0)}$ at the free surface. As shown in Fig. 3(b), at the free surface ($z = z^{(0)} = 0$), $\hat{\mathbf{R}}_{ud}^{(0)}$ is defined as:

$$\mathbf{C}_d^{(1)} = \hat{\mathbf{R}}_{ud}^{(0)}\mathbf{C}_u^{(1)}. \tag{14}$$

Applying the free surface boundary condition (i.e. $\mathbf{y}_2^{(1)}(z^{(0)}) = [\boldsymbol{\tau}_S^{(1)}(z^{(0)})\ \boldsymbol{\tau}_R^{(1)}(z^{(0)})]^{\mathrm{T}} = \mathbf{0}$) to eq. (9), we have:

$$\mathbf{E}_{21}^{(1)}\mathbf{C}_d^{(1)} + \mathbf{E}_{22}^{(1)}\mathbf{\Lambda}_u^{(1)}(z^{(0)})\mathbf{C}_u^{(1)} = \mathbf{0}. \tag{15}$$

Submitting eq. (14) to eq. (15), we can obtain:

$$\hat{\mathbf{R}}_{ud}^{(0)} = -\left(\mathbf{E}_{21}^{(1)}\right)^{-1}\mathbf{E}_{22}^{(1)}\mathbf{\Lambda}_u^{(1)}(z^{(0)}). \tag{16}$$

Thus, $\hat{\mathbf{T}}_u^{(j)}$ and $\hat{\mathbf{R}}_{ud}^{(j)}$ can be calculated by eqs (13) and (16).

For downgoing waves, as shown in Fig. 3(c), the generalized reflection and transmission coefficients ($\hat{\mathbf{T}}_d^{(j)}$ and $\hat{\mathbf{R}}_{du}^{(j)}$) at the layer

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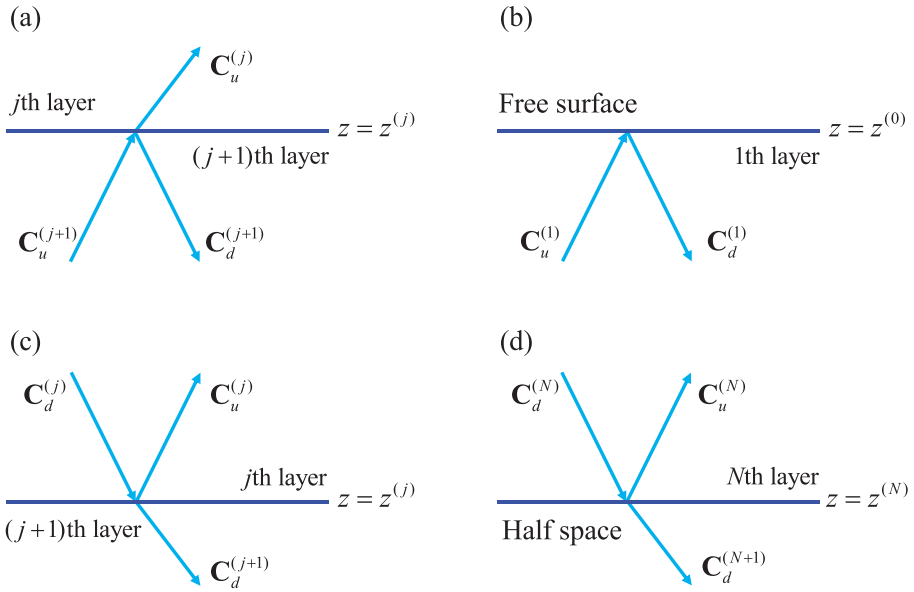


Figure 3. Schematic diagram of physical interpretation of generalized reflection and transmission. (a) upgoing waves at the layer interface; (b) upgoing waves at the free surface; (c) downgoing waves at the layer interface and (d) downgoing waves at half-space.

interface $z = z^{(j)}$ are defined as:

$$\begin{cases} \mathbf{C}_d^{(j+1)} = \hat{\mathbf{T}}_d^{(j)} \mathbf{C}_d^{(j)} \\ \mathbf{C}_u^{(j)} = \hat{\mathbf{R}}_{du}^{(j)} \mathbf{C}_d^{(j)} \end{cases}. \tag{17}$$

Based on the continuity boundary conditions, submitting eq. (17) to eq. (11), we can derive:

$$\begin{pmatrix} \mathbf{\Lambda}_d^{(j)}(z^{(j)}) \\ \hat{\mathbf{R}}_{du}^{(j)} \end{pmatrix} \mathbf{C}_d^{(j)} = \begin{pmatrix} \mathbf{W}_{11}^{(j)} & \mathbf{W}_{12}^{(j)} \\ \mathbf{W}_{21}^{(j)} & \mathbf{W}_{22}^{(j)} \end{pmatrix} \begin{pmatrix} \mathbf{I} \\ \mathbf{\Lambda}_u^{(j+1)}(z^{(j)}) \hat{\mathbf{R}}_{du}^{(j+1)} \end{pmatrix} \hat{\mathbf{T}}_d^{(j)} \mathbf{C}_d^{(j)}, \tag{18}$$

where $\mathbf{W}^{(j)} = (\mathbf{E}^{(j)})^{-1} \mathbf{E}^{(j+1)} = (\mathbf{O}^{(j)})^{-1}$, and $\mathbf{W}_{11}^{(j)}$, $\mathbf{W}_{12}^{(j)}$, $\mathbf{W}_{21}^{(j)}$ and $\mathbf{W}_{22}^{(j)}$ are its submatrices. Further organizing eq. (18), the formulae of $\hat{\mathbf{T}}_d^{(j)}$ and $\hat{\mathbf{R}}_{du}^{(j)}$ can be obtained:

$$\begin{cases} \hat{\mathbf{T}}_d^{(j)} = \left[\mathbf{W}_{11}^{(j)} + \mathbf{W}_{12}^{(j)} \mathbf{\Lambda}_u^{(j+1)}(z^{(j)}) \hat{\mathbf{R}}_{du}^{(j+1)} \right]^{-1} \mathbf{\Lambda}_d^{(j)}(z^{(j)}) \\ \hat{\mathbf{R}}_{du}^{(j)} = \left[\mathbf{W}_{21}^{(j)} + \mathbf{W}_{22}^{(j)} \mathbf{\Lambda}_u^{(j+1)}(z^{(j)}) \hat{\mathbf{R}}_{du}^{(j+1)} \right] \hat{\mathbf{T}}_d^{(j)} \end{cases}, \tag{19}$$

where $j = N, N-1, \ldots, 2, 1$. According to eq. (19), to calculate $\hat{\mathbf{T}}_d^{(j)}$ and $\hat{\mathbf{R}}_{du}^{(j)}$ for the arbitrary layer interface, we should first know the reflection coefficient $\hat{\mathbf{R}}_{du}^{(N+1)}$. As Fig. 3(d) shows, at the layer interface $z = z^{(N)}$, $\hat{\mathbf{R}}_{du}^{(N+1)}$ can be defined as:

$$\mathbf{C}_u^{(N+1)} = \hat{\mathbf{R}}_{du}^{(N+1)} \mathbf{C}_d^{(N+1)}. \tag{20}$$

According to the radiation boundary condition (i.e. $\mathbf{y}^{(N+1)}(z) \to \mathbf{0}$, $z \to +\infty$) which indicates that only downgoing waves exist in half-space, we have:

$$\begin{cases} \mathbf{C}_d^{(N+1)} \neq \mathbf{0} \\ \mathbf{C}_u^{(N+1)} = \mathbf{0} \end{cases}. \tag{21}$$

Submitting eq. (21) to eq. (20), we can obtain:

$$\hat{\mathbf{R}}_{du}^{(N+1)} = \mathbf{0}. \tag{22}$$

Consequently, through eqs (19) and (22), $\hat{\mathbf{T}}_d^{(j)}$ and $\hat{\mathbf{R}}_{du}^{(j)}$ can be calculated.

The above derivation process is also applicable to the SH wave case. For SH waves, the general solution (eq. B12) of the ordinary differential equations can be expressed as:

$$\mathbf{y}^{(j)}(z) = \begin{pmatrix} E_{11}^{(j)} & E_{12}^{(j)} \\ E_{21}^{(j)} & E_{22}^{(j)} \end{pmatrix} \begin{pmatrix} \Lambda_d^{(j)}(z) & \\ & \Lambda_u^{(j)}(z) \end{pmatrix} \begin{pmatrix} C_d^{(j)} \\ C_u^{(j)} \end{pmatrix}, \tag{23}$$

where $E_{11}^{(j)}$, $E_{12}^{(j)}$, $E_{21}^{(j)}$ and $E_{22}^{(j)}$ are elements of matrix $\mathbf{E}^{(j)}$; $\Lambda_d^{(j)}(z)$ and $\Lambda_u^{(j)}(z)$ are elements of matrix $\mathbf{\Lambda}^{(j)}$; $C_d^{(j)} = C_{SHd}^{(j)}$ and $C_u^{(j)} = C_{SHu}^{(j)}$ are elements of matrix $\mathbf{C}^{(j)}$. The explicit expressions of $\mathbf{E}^{(j)}$and $\mathbf{\Lambda}^{(j)}$ are given in Appendix C. For upgoing waves, the generalized reflection and transmission coefficient ($\hat{T}_u^{(j)}$ and $\hat{R}_{ud}^{(j)}$) can be calculated by:

$$\begin{cases} \hat{T}_u^{(j)} = \left[O_{21}^{(j)} \Lambda_d^{(j)}(z^{(j)}) \hat{R}_{ud}^{(j-1)} + O_{22}^{(j)} \right]^{-1} \Lambda_u^{(j+1)}(z^{(j)}) \\ \hat{R}_{ud}^{(j)} = \left[O_{11}^{(j)} \Lambda_d^{(j)}(z^{(j)}) \hat{R}_{ud}^{(j-1)} + O_{12}^{(j)} \right] \hat{T}_u^{(j)} \\ \hat{R}_{ud}^{(0)} = -\left(E_{21}^{(1)} \right)^{-1} E_{22}^{(1)} \Lambda_u^{(1)}(z^{(0)}) \end{cases}, \tag{24}$$

where $j = 1, 2, \ldots, N-2, N-1$; $\mathbf{O}^{(j)} = (\mathbf{E}^{(j+1)})^{-1} \mathbf{E}^{(j)}$, and $O_{11}^{(j)}$, $O_{12}^{(j)}$, $O_{21}^{(j)}$ and $O_{22}^{(j)}$ are its elements. Similarly, for downgoing waves, the generalized reflection and transmission coefficient ($\hat{T}_d^{(j)}$ and $\hat{R}_{du}^{(j)}$) can be calculated by:

$$\begin{cases} \hat{T}_d^{(j)} = \left[W_{11}^{(j)} + W_{12}^{(j)} \Lambda_u^{(j+1)}(z^{(j)}) \hat{R}_{du}^{(j+1)} \right]^{-1} \Lambda_d^{(j)}(z^{(j)}) \\ \hat{R}_{du}^{(j)} = \left[W_{21}^{(j)} + W_{22}^{(j)} \Lambda_u^{(j+1)}(z^{(j)}) \hat{R}_{du}^{(j+1)} \right] \hat{T}_d^{(j)} \\ \hat{R}_{du}^{(N+1)} = 0 \end{cases}, \tag{25}$$

where $j = N, N-1, \ldots, 2, 1$; $\mathbf{W}^{(j)} = (\mathbf{E}^{(j)})^{-1} \mathbf{E}^{(j+1)} = (\mathbf{O}^{(j)})^{-1}$, and $W_{11}^{(j)}$, $W_{12}^{(j)}$, $W_{21}^{(j)}$ and $W_{22}^{(j)}$ are its elements.

2.3 A family of secular functions

After having derived the generalized R/T coefficients, to calculate the modal solutions of surface waves in viscoelastic-VTI media,

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we need to further introduce the corresponding family of secular functions (He *et al.* 2006; Wu & Chen 2016).

Let's first consider the *P*-SV wave case. Extracting the definition formula for the generalized reflection coefficients in the *j*th layer medium from eqs (10) and (17), we have:

$$\begin{cases} \mathbf{C}_d^{(j)} = \hat{\mathbf{R}}_{ud}^{(j-1)} \mathbf{C}_u^{(j)} \\ \mathbf{C}_u^{(j)} = \hat{\mathbf{R}}_{du}^{(j)} \mathbf{C}_d^{(j)} \end{cases}, \tag{26}$$

where $j = 1, 2, \ldots, N-1, N$. Further organizing eq. (26), we have:

$$\left(\mathbf{I} - \hat{\mathbf{R}}_{ud}^{(j-1)} \hat{\mathbf{R}}_{du}^{(j)} \right) \mathbf{C}_d^{(j)} = \mathbf{0}. \tag{27}$$

The condition for the existence of non-zero $\mathbf{C}_d^{(j)}$ in eq. (27) are as follows:

$$\det \left\{ \mathbf{I} - \hat{\mathbf{R}}_{ud}^{(j-1)} \hat{\mathbf{R}}_{du}^{(j)} \right\} = 0, \tag{28}$$

where $\det\{\cdot\}$ represents the calculation of the matrix determinant. Eq. (28) denotes the dispersion equations corresponding to P-SV waves in various layers above the half-space. In addition, submitting $\mathbf{C}_u^{(1)} = \hat{\mathbf{R}}_{du}^{(1)} \mathbf{C}_d^{(1)}$ to eq. (24) and considering the condition for the existence of non-zero $\mathbf{C}_d^{(1)}$, we can obtain another dispersion equation related to the free surface boundary:

$$\det \left\{ \mathbf{E}_{21}^{(1)} + \mathbf{E}_{22}^{(1)} \mathbf{\Lambda}_u^{(1)}(z^{(0)}) \hat{\mathbf{R}}_{du}^{(1)} \right\} = 0. \tag{29}$$

According to eqs (28) and (29), we define the family of secular functions of *P*-SV waves in viscoelastic-VTI media:

$$F_j^R(\omega, k, M_{11}, M_{13}, M_{33}, M_{55}) = \begin{cases} \det \left\{ \mathbf{E}_{21}^{(1)} + \mathbf{E}_{22}^{(1)} \mathbf{\Lambda}_u^{(1)}(z^{(0)}) \hat{\mathbf{R}}_{du}^{(1)} \right\}, \text{for} j = 0 \\ \det \left\{ \mathbf{I} - \hat{\mathbf{R}}_{ud}^{(j-1)} \hat{\mathbf{R}}_{du}^{(j)} \right\}, \text{for} j = 1, 2, \ldots, N \end{cases}. \tag{30}$$

As eq. (30) shows, the family of secular functions of *P*-SV waves, which consist of $N+1$ secular functions, are the functions of angular frequency ω, horizontal complex wavenumber k and four independent complex moduli (i.e. M_{11}, M_{13}, M_{33} and M_{55}). Through solving the dispersion equations $F_j^R(\omega, k, M_{11}, M_{13}, M_{33}, M_{55}) = 0$, the modal solutions (i.e. k) of Rayleigh waves can be obtained.

Similarly, we can also derive the family of secular functions of SH waves in viscoelastic-VTI media as follows:

$$F_j^L(\omega, k, M_{55}, M_{66}) = \begin{cases} E_{21}^{(1)} + E_{22}^{(1)} \Lambda_u^{(1)}(z^{(0)}) \hat{R}_{du}^{(1)}, \text{for} j = 0 \\ 1 - \hat{R}_{ud}^{(j-1)} \hat{R}_{du}^{(j)}, \text{for} j = 1, 2, \ldots, N \end{cases}. \tag{31}$$

From eq. (31) we see that the family of secular functions of SH waves are the functions of angular frequency ω, horizontal complex wavenumber k, and two independent complex moduli (i.e. M_{55} and M_{66}). The modal solutions (i.e. k) of Love waves can be obtained by solving the dispersion equations $F_j^L(\omega, k, M_{55}, M_{66}) = 0$.

However, in practice, Wu & Chen (2016, 2022) have proven by many examples that the family of secular functions is not all required. They pointed out that only the secular function (i.e. F_0^R or F_0^L) associated with the free surface and those associated with all of the low-velocity layers need to be invoked to accurately and efficiently find all of the physically existent modes at any given frequency. In addition, once the modal solutions (i.e. k) of surface waves are determined, we can easily determine the unknown wave amplitude vector $\mathbf{C}^{(j)}$, and further calculate the corresponding eigenfunctions including eigendisplacements and eigentractions through eq. (B12) (see Appendix D), as done by Chen (1993). Through the determined horizontal and vertical eigendisplacements of Rayleigh waves, we can also calculate the theoretical horizontal-to-vertical spectral ratios (HVSR) curves (i.e. ellipticity) of Rayleigh waves (see Appendix E).

3 ALGORITHM IMPLEMENTATION AND VALIDATION

3.1 Algorithm implementation

Compared with the case of isotropic-elastic media, the stiffness matrix in viscoelastic-VTI media contains more independent components, and they are complex and frequency-dependent. Moreover, the surface-wave modal solutions (i.e. the zero points of secular functions or the roots of dispersion equations) also become complex. Therefore, it is not possible to directly use the root-finding method in the real domain to calculate the zero points of secular functions. Instead, it is necessary to find the complex roots, namely the complex wavenumbers (k) of surface waves, in the complex domain.

Considering the characteristic that the zero points of the secular functions are also extreme points, we can design a numerical scheme of combining the root-searching method with the local optimization inversion method to calculate the complex wavenumbers of surface waves in viscoelastic-VTI media. Inversion reflects the process of determining a model through data fitting. In this study, the secular function values are used as the data item (where the zero values are used as the known data item), and the real and imaginary parts (i.e. k_r and k_i) of the surface-wave complex wavenumbers ($k = k_r + ik_i$) are regarded as the model item to be inverted. The core workflow mainly includes the following four steps:

Step 1 is to calculate the approximate k_r in the real domain. For isotropic-viscoelastic media, Yuan *et al.* (2018) introduced a strategy to approximately calculate the phase-velocity (c_r) dispersion curves of surface waves in the real domain, that is, replacing the complex velocities of *P* and *S* waves in the dispersion equation with the corresponding phase velocities. According to eqs (4) and (5), the modulus $M_{p,mn}^{(j)}$ corresponding to phase velocity can be expressed as:

$$M_{p,mn}^{(j)}(\omega) = \rho^{(j)} \left(V_{p,mn}^{(j)}(\omega) \right)^2. \tag{32}$$

Therefore, by replacing the complex modulus (M_{mn}) in eqs (30) and (31) with the real modulus ($M_{p,mn}$), the approximate phase-velocity dispersion curves of surface waves in viscoelastic-VTI media can be directly calculated by the root-finding method in the real domain (e.g. bisection method). The root-searching process based on the family of secular functions is similar to the works of Wu & Chen (2016, 2022), which are aimed at the isotropic-elastic media. This step can output the approximate k_r (by $k_r = \omega / c_r$) and the used secular-function type for each determined root, and this prior information will be used in the following steps. It is worth noting that the obtained prior information is better than that obtained directly from elastic media, which will be discussed in Section 3.2.

Step 2 is to update k_i in the complex domain. For each given frequency and mode, based on the determined k_r and secular-function type in step 1, we can iteratively update the corresponding k_i in the complex domain through a local optimization method (e.g. damped least squares method). In order from high frequency to low frequency, the initial values of k_i can be taken as $\mathrm{Im}[\omega / V_{66}^{(1)}(\omega)]$ (i.e. the imaginary part of the *S*-wave complex wavenumber in the first

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layer medium) for the first frequency point, and then for other frequency points, they can be set to the results of previous frequency point on the same surface-wave modes.

Step 3 is to update k_r in the complex domain. For each given frequency and mode, based on the determined k_i in step 2 and the secular-function type in step 1, the corresponding k_r can also be iteratively updated in the complex domain through a local optimization method.

Step 4 is to further update k_i and k_r by repeating steps 2 and 3, but the corresponding initial values are set to the updated results in the previous step. Usually, by repeating this process once, we can obtain the complex wavenumbers ($k = k_r + ik_i$) of surface waves meeting the accuracy requirement. Finally, considering the relationship $k = \omega/c_r - i\alpha$, the phase-velocity dispersion curves ($c_r(\omega)$) and attenuation coefficient curves ($\alpha(\omega)$) of surface waves in viscoelastic-VTI media can be determined by the following formulas:

$$c_r(\omega) = \omega \big/ \mathrm{Re}\,[k(\omega)], \tag{33}$$

$$\alpha(\omega) = -\mathrm{Im}\,[k(\omega)]\,. \tag{34}$$

Following the methodology in Section 2 and the above steps, we can achieve the computation program of the surface-wave modal solutions in viscoelastic-VTI media. Using this program, we can also obtain the results in isotropic-viscoelastic media by setting $\varepsilon = \delta = \gamma = 0$ (see eq. 7) and $\varepsilon_Q = \delta_Q = \gamma_Q = 0$ (see eq. 8), the results in anisotropic-elastic media by setting $\upsilon_{mn} = 0$ (equivalent to $Q_{mn} = +\infty$; see eq. 3), as well as the results in isotropic-elastic media by simultaneously setting $\varepsilon = \delta = \gamma = 0$, $\varepsilon_Q = \delta_Q = \gamma_Q = 0$ and $\upsilon_{mn} = 0$. For the cases in elastic media, only step 1 needs to be executed in the above steps, due to the zero attenuation coefficient.

3.2 Algorithm validation

To test our algorithm, we adopt two relatively typical six-layer half-space models, i.e. Models 1 and 2. Model 1 can reflect the Earth's internal structure at the thick crustal scale with weaker attenuation (larger Q values), while Model 2 conforms to the near-surface structure of the Earth with stronger attenuation (smaller Q values; e.g. near-surface sedimentary layers). Tables 1 and 2 show the model parameters of viscoelastic-VTI media of Models 1 and 2 used for the case of P-SV waves, respectively, and those used for the case of SH waves are displayed in Tables S1 and S2. For simplification, in this study, the relationships of $Q_{S0} = V_{S0}/10$, $Q_{P0} = 2Q_{S0}$, $\delta = \varepsilon/2$, $\varepsilon_Q = \varepsilon$, $\gamma_Q = \gamma$ and $\delta_Q = \delta$ are assumed in model parameters, and the reference frequency f_r in the viscoelasticity mechanism (eq. 3) is set to 0.2 and 25 Hz for Models 1 and 2, respectively.

To verify the correctness of our algorithm, we benchmark the surface-wave modal solutions (i.e. k) based on our GRTM against those from other methods. Currently, there have been no reports on the calculation method of modal solutions of surface waves in viscoelastic-VTI media. Thus, by replacing the elasticity constants (i.e. $C_{mn}^{(j)}$, see eq. 7) with the corresponding complex moduli (i.e. $M_{mn}^{(j)}$, see eq. 3), we extend the dispersion equation of Rayleigh waves derived by the reduced delta matrix method (Ikeda & Matsuoka 2013) and that of Love waves developed by the Thomson–Haskell method (Anderson 1961) from the elastic-VTI media to the viscoelastic-VTI media. Then, they can also be used to calculate the modal solutions of surface waves in viscoelastic-VTI media by following the implementation steps in section 3.1. Consequently, we adopt these two methods to validate our algorithm.

Using the models in Tables 1 and 2 (i.e. Models 1 and 2), we calculate the modal solutions of Rayleigh waves in four different media by different methods, respectively. Figs 4 and 5 display the comparisons of the Rayleigh-wave modal solutions between our GRTM (red solid lines) and the reduced delta matrix method (labeled as RDMM; blue points) for Models 1 and 2, respectively. As Figs 4 and 5 show, whether the complex modal solutions for isotropic-viscoelastic (IV) and anisotropic-viscoelastic (AV) media in the frequency-phase velocity-attenuation coefficient domain (Figs 4b, d, 5b and d), or the real modal solutions for isotropic-elastic (IE) and anisotropic-elastic (AE) in the frequency-phase velocity domain (Figs 4a, c, 5a and c), all results of GRTM can well fit with those of RDMM. Similar verification for Love waves is also performed by the comparisons of the modal solutions between our GRTM and the Thomson–Haskell method (see Text S2 as well as Figs S1 and S2). Thus, the excellent comparisons confirm the correctness of our algorithm.

In addition, from Figs 4 and 5, we can also find that the surface-wave modal solutions in viscoelastic media are distributed in the 3-D space of the frequency-phase velocity-attenuation coefficient domain, and for any given frequency and modal phase velocity, there is only one corresponding modal attenuation coefficient, which can be uniquely determined by local optimization. Therefore, in Section 3.1, step 1 is important for determining the modal attenuation coefficients in the next step, because it can provide very good initial results of the modal phase velocities and these results are better than those obtained from elastic media. As already confirmed above (see Figs 4 and 5), the strategy is valid not only for deep layered earth models with weak attenuation (e.g. Model 1), but also for shallow layered earth models with stronger attenuation (e.g. Model 2).

To explain this point, taking Model 2 as an example, in Fig. 6, we compare the phase-velocity dispersion curves of Rayleigh waves in IV and AV media approximately calculated by only performing step 1 (blue dashed–dotted lines in Figs 6a and b) with those obtained by executing all steps (regarded as accurate solutions; red solid lines in Figs 6a and b), and also compare the accurate results in IV and AV media (red solid lines in Figs 6c and d) with those in IE and AE media (blue dashed–dotted lines in Figs 6c and d), respectively. As depicted in Fig. 6, whether in IV media or AV media, the results obtained from Step 1 (blue dashed–dotted lines in Figs 6a and b) significantly better approximate the accurate solutions (red solid lines) compared to those derived from corresponding elastic media (blue dashed–dotted lines in Figs 6c and d), particularly for higher-order modes, which serves to confirm the aforementioned assertion. Similar comparisons for Love waves can also lead to the same conclusion (see Fig. S3). Moreover, it also explains that the approximate results from step 1 can be used for the study on modelling and inversion of phase-velocity dispersion curves of surface waves in viscoelastic media in general.

4 ANALYSIS OF SURFACE WAVE CHARACTERISTICS

After achieving the computation of the surface-wave modal solutions, we can further analyse the dispersion and attenuation characteristics of both Rayleigh and Love waves by comparing the modelling results of different media. Focusing on investigating the effects of medium properties on surface waves, we design a relatively simple model, i.e. the four-layer half-space model with viscoelastic-VTI media (Model 3), to avoid interference from complex geological structures. The parameters describing the velocity anisotropy

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Table 1. Parameters of Model 1 used for the *P*-SV wave case.

Layer no.	z (km)	ρ (g cm^{-3})	V_{S0} (m s^{-1})	V_{P0} (m s^{-1})	Q_{S0}	Q_{P0}	ε	δ	ε_Q	δ_Q
1	0–10	2.5	2500	4500	250	500	0.10	0.05	0.10	0.05
2	10–20	2.6	3300	6100	330	660	0.12	0.06	0.12	0.06
3	20–30	2.7	3600	6300	360	720	0.14	0.07	0.14	0.07
4	30–40	2.8	4000	6600	400	800	0.16	0.08	0.16	0.08
5	40–50	2.9	4300	6800	430	860	0.18	0.09	0.18	0.09
6	50 to $+\infty$	3.0	4500	7200	450	900	0.20	0.10	0.20	0.10

Table 2. Parameters of Model 2 used for the *P*-SV wave case.

Layer no.	z (m)	ρ (g cm^{-3})	V_{S0} (m s^{-1})	V_{P0} (m s^{-1})	Q_{S0}	Q_{P0}	ε	δ	ε_Q	δ_Q
1	0–3	1.82	190	800	19	38	0.10	0.05	0.10	0.05
2	3–7	1.86	320	1200	32	64	0.16	0.08	0.16	0.08
3	7–12	1.91	280	1000	28	56	0.14	0.07	0.14	0.07
4	12–17	1.95	460	1400	46	92	0.20	0.10	0.20	0.10
5	17–23	2.0	630	1900	63	126	0.22	0.11	0.22	0.11
6	23 to $+\infty$	2.1	750	2300	75	150	0.24	0.12	0.24	0.12

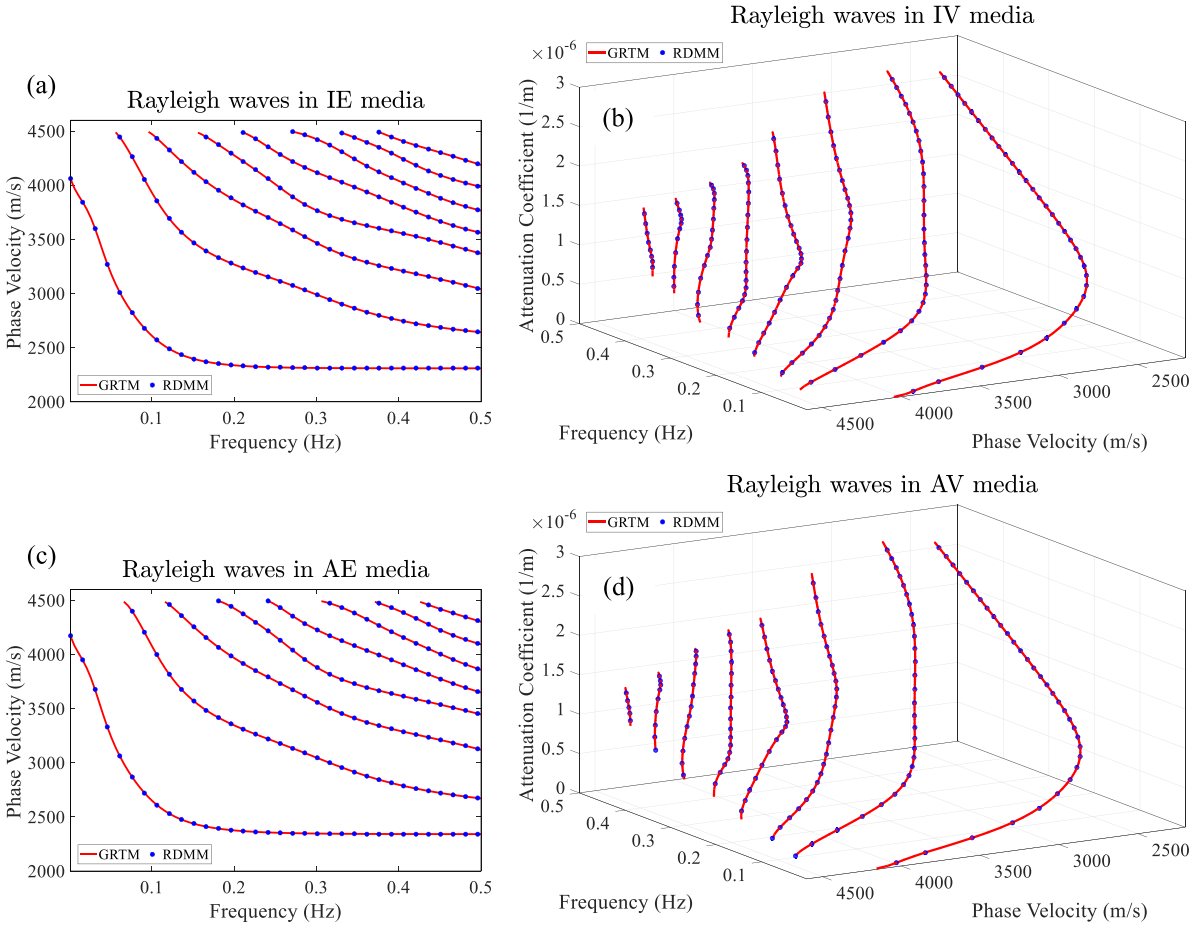


Figure 4. Comparisons of Rayleigh-wave modal solutions between our GRTM (red solid lines) and the reduced delta matrix method (labeled as RDMM; blue points) for Model 1 in (a) isotropic-elastic (IE), (b) isotropic-viscoelastic (IV), (c) anisotropic-elastic (AE) and (d) anisotropic-viscoelastic (AV) media.

(ε, δ and γ) and attenuation anisotropy (ε_Q, δ_Q and γ_Q) are set to be the same in each layer of the model. The model parameters used for the cases of *P*-SV and SH waves are displayed in Tables 3 and 4, respectively, and they accord with the near-surface sedimentary geological environment. The reference frequency f_r is set to 25 Hz in the viscoelasticity mechanism (eq. 3). As introduced in section 3.1, using the computation program of surface-wave modal solutions based on the theory of viscoelastic-VTI media, we can also simulate the results in isotropic-elastic (IE), anisotropic-elastic (AE), isotropic-viscoelastic (IV) and anisotropic-viscoelastic (AV) media by resetting the corresponding parameters, respectively.

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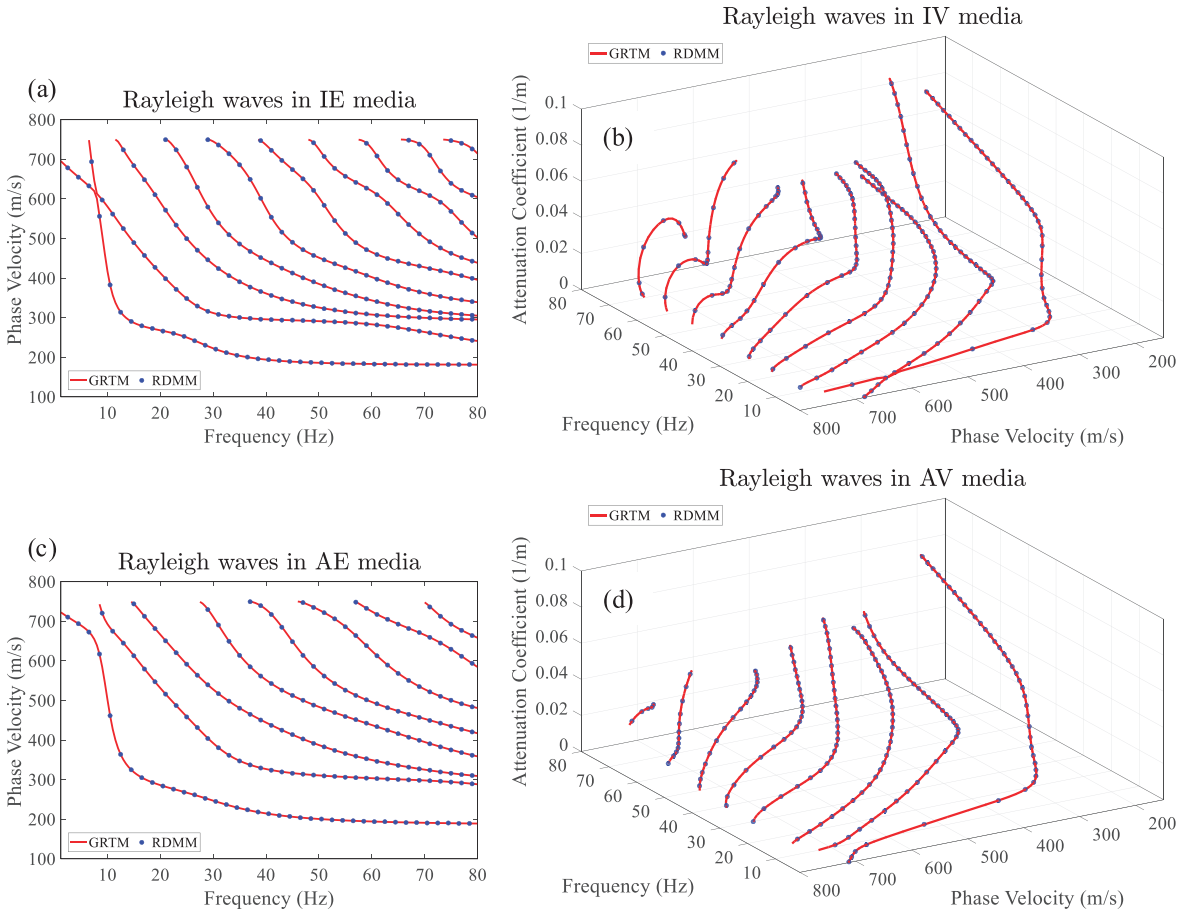


Figure 5. Comparisons of Rayleigh-wave modal solutions between our GRTM (red solid lines) and the reduced delta matrix method (labeled as RDMM; blue points) for Model 2 in (a) isotropic-elastic (IE), (b) isotropic-viscoelastic (IV), (c) anisotropic-elastic (AE) and (d) anisotropic-viscoelastic (AV) media.

Figs 7(a) and (b) show the modal solutions of Rayleigh waves in IV and AV media, respectively. They are distributed in the 3-D space of the frequency-phase velocity-attenuation coefficient domain. Their projections to the frequency-phase velocity domain are usually known as the phase-velocity dispersion curves (Figs 8c and d), while those to the frequency-attenuation coefficient domain are usually called the attenuation coefficient curves (Figs 7c and d). Because of the zero attenuation coefficient in elastic media, the modal solutions of Rayleigh waves in IE and AE media are distributed in the 2-D space of the frequency-phase velocity domain, as the phase-velocity dispersion curves displayed in Figs 8(a) and (b). Similarly, the modal solutions of Love waves in four different media are shown in Figs 9 and 10. Fig. 11 shows the phase-velocity differences of surface-wave dispersion curves in IV, AE and AV media to those in IE media. In this study, the methodology of viscoelastic-VTI media comprises the properties of velocity anisotropy, viscoelasticity, and attenuation anisotropy. The phase-velocity differences between AE and IE media (Figs 11c and d) as well as between IV and IE media (Figs 11a and b) reveal the effects of velocity anisotropy and viscoelasticity on surface waves, respectively, while those between AV and IE media (Figs 11e and f) embody the superimposed effects of these three properties. In the following text, we will explore the effects of these properties on the dispersion and attenuation characteristics of surface waves in detail by comparing the results of different media.

4.1 Effects of velocity anisotropy

The velocity anisotropy of TI media is described by three dimensionless Thomsen's parameters (i.e. ε, δ and γ). Parameters ε and δ are used for controlling the effects of velocity anisotropy on Rayleigh waves, while parameter γ is aimed at the case of Love waves. By comparing the modal solutions in AE and IE media, we can analyse the effects of velocity anisotropy on the dispersion characteristic of surface waves. We first analyse the case of Rayleigh waves.

As shown in the first row of Fig. 8, we can find that for both AE and IE media, the cut-off phase velocities of all Rayleigh-wave higher modes in the low-frequency end are equal to the SV-wave velocity in the half-space layer (i.e. $V_{SV}^{(4)}(90^\circ) = V_{S0}^{(4)} = 500\ \mathrm{m\,s^{-1}}$), while the phase velocity of the fundamental mode in the high-frequency end approaches 98 per cent of the SV-wave velocity in the first layer (i.e. $0.98V_{SV}^{(1)}(90^\circ) = 0.98V_{S0}^{(1)} \approx 176\ \mathrm{m\,s^{-1}}$). Thus, it explains that the velocity anisotropy of the VTI medium will not change the phase-velocity distribution range of Rayleigh waves. As is well known, Rayleigh waves are the result of interfering P and SV waves on the free surface, and their phase velocities are most sensitive to the SV-wave velocities (Aki & Richards 1980). As eqs (A8) and (A9) display, the velocity of SV waves in VTI medium in the pure mode direction is equal to that in the corresponding isotropic medium, for example $V_{SV}(0^\circ) = V_{S0}$, and $V_{SV}(90^\circ) = V_{S0}$, which may be the reason for the above phenomenon.

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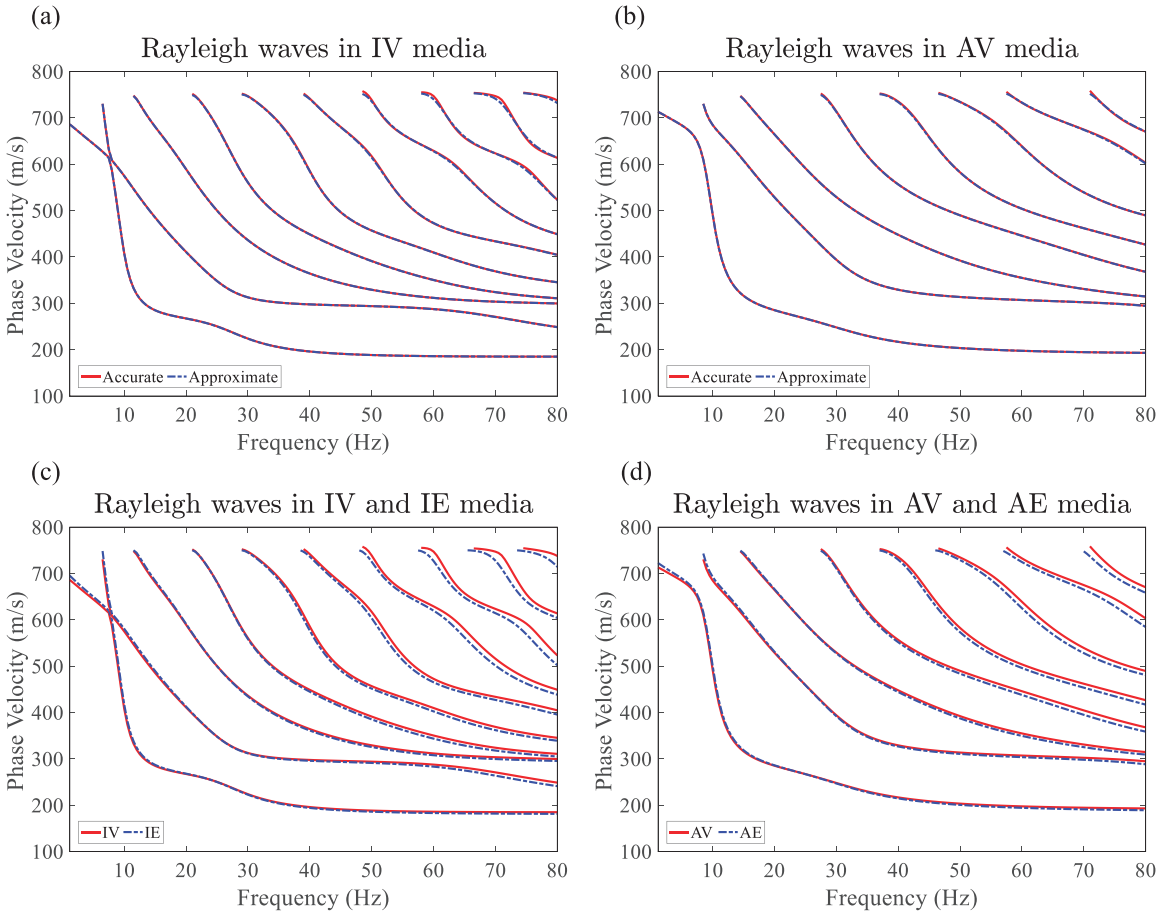


Figure 6. Comparisons of phase-velocity dispersion curves of Rayleigh waves for Model 2: (a, b) comparing the accurate solutions (red solid lines) with approximate results (blue dashed–dotted lines) in isotropic-viscoelastic (IV) and anisotropic-viscoelastic (AV) media; (c, d) comparing the accurate results in IV and AV media (red solid lines) with those in IE and AE media (blue dashed–dotted lines).

Table 3. Parameters of Model 3 used for the *P*-SV wave case.

Layer no.	z (m)	ρ (g cm^{-3})	V_{S0} (m s^{-1})	V_{P0} (m s^{-1})	Q_{S0}	Q_{P0}	ε	δ	ε_Q	δ_Q
1	0–5	2.0	180	800	18	36	0.1	0.05	0.1	0.05
2	5–10	2.0	300	1200	30	60	0.1	0.05	0.1	0.05
3	10–15	2.0	420	1500	42	84	0.1	0.05	0.1	0.05
4	15 to $+\infty$	2.0	500	1800	50	100	0.1	0.05	0.1	0.05

Table 4. Parameters of Model 3 used for the SH-wave case.

Layer no.	z (m)	ρ (g cm^{-3})	V_{S0} (m s^{-1})	Q_{S0}	γ	γ_Q
1	0–5	2.0	180	18	0.1	0.1
2	5–10	2.0	300	30	0.1	0.1
3	10–15	2.0	420	42	0.1	0.1
4	15 to $+\infty$	2.0	500	50	0.1	0.1

In addition, there are also significant differences caused by the velocity anisotropy of the VTI medium. Within the same frequency-phase velocity window range, compared with the results of IE media (Fig. 8a), for the phase-velocity dispersion curves of Rayleigh waves in AE media (Fig. 8b), the higher modes present higher cut-off frequency and narrower frequency-band range, and the phase-velocity value at the same mode and frequency becomes larger. It seems the dispersion curves of Rayleigh waves in AE media are stretched towards the high-frequency end as a whole with different degrees. This phenomenon may be related to the velocities of quasi-SV waves in the non-pure mode direction [$\theta \neq m \cdot 90^\circ$ (here, m is any integer) in eq. A5] in each layer. As Fig. 2 shows, they are greater than the velocities of SV waves in isotropic medium of the corresponding layer. Consequently, Rayleigh waves with higher phase velocities (c_r) have higher frequencies (f) under detecting the same depth (i.e. having the same wavelength (λ); $\lambda = c_r/f$), which is consistent with the above phenomenon. To better reflect their differences, in Fig. 11(c), we show the phase velocity differences of Rayleigh-wave dispersion curves in AE media to those in IE media. As Fig. 11(c) displays, the phase velocity differences of all modes are much greater than zero values, and roughly distributed between 5 and 95 m s^{-1} (the increasing ratio is between 3 and 30 per cent); the degree of increase varies among different modes, and the higher modes (M1–M6) present a greater increase than the fundamental mode (M0).

The effects of velocity anisotropy on Love waves are obviously different from those on Rayleigh waves. As the first row of Fig. 10 shows, compared with the results in IE media (Fig. 10a),

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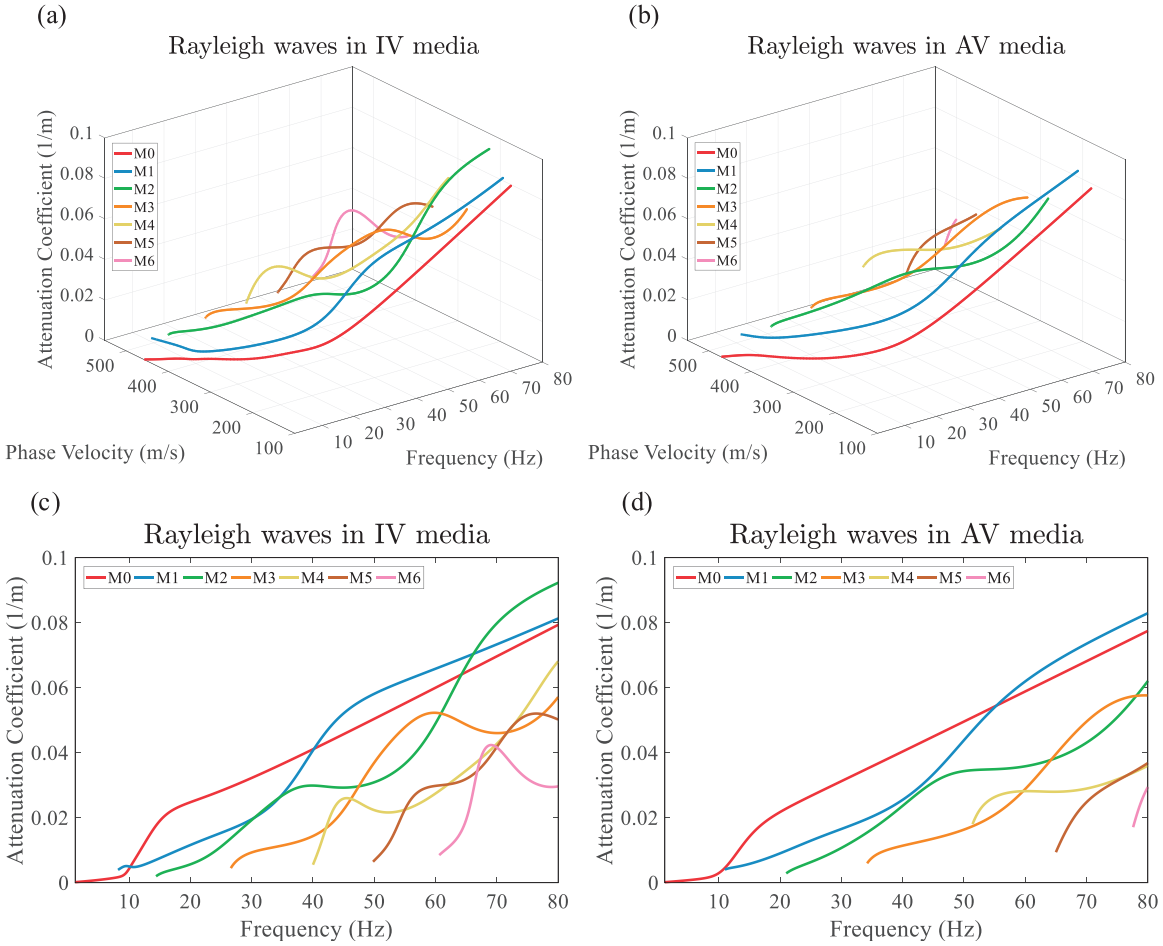


Figure 7. The Rayleigh-wave modal solutions in (a, c) isotropic-viscoelastic (IV) and (b, d) anisotropic-viscoelastic (AV) media. (a) and (b) correspond to the results in the frequency-phase velocity-attenuation coefficient domain; (c) and (d) correspond to their projections to the frequency-attenuation coefficient domain, that is the attenuation coefficient curves. M0, M1, M2, M3, M4, M5 and M6 represent the fundamental, first-higher, second-higher, third-higher, fourth-higher, fifth-higher and sixth-higher modes, respectively.

for the phase-velocity dispersion curves of Love waves in AE media (Fig. 10b), the frequency bands of all modes remain unchanged, but their phase-velocity distribution ranges are significantly changed. The modal information of Love waves in IE media is distributed between the velocity of SH waves in the first layer (i.e. $V_{S0}^{(1)} = 180\ \mathrm{m\,s^{-1}}$) and that in the half-space layer (i.e. $V_{S0}^{(4)} = 500\ \mathrm{m\,s^{-1}}$). However, in AE media, the modal information of Love waves has a broader phase-velocity distribution range with the lower bound $V_{SH}^{(1)}(90^\circ) = V_{S0}^{(1)}\sqrt{1+2\gamma^{(1)}} \approx 197\ \mathrm{m\,s^{-1}}$ and the upper bound $V_{SH}^{(4)}(90^\circ) = V_{S0}^{(4)}\sqrt{1+2\gamma^{(4)}} \approx 548\ \mathrm{m\,s^{-1}}$. Therefore, we can conclude that the velocity anisotropy of the VTI medium will cause a change in the phase-velocity distribution range of Love waves. It can be the reason that Love waves are formed by the constructive interference of multiple reflections of SH waves, while the velocity of SH waves in VTI medium in the horizontal direction is not equal to that in the corresponding isotropic medium (i.e. $V_{SH}(90^\circ) = V_{S0}\sqrt{1+2\gamma}$ in eq. A9). Moreover, for any given mode and frequency, Love waves in AE media present a greater phase-velocity value, as the phase velocity differences of Love-wave dispersion curves between AE and IE media show (Fig. 11d). The phase velocity differences of all modes are greater than zero values and decrease regularly with increasing frequency. Interestingly, for any given mode and frequency, we find that the increasing ratio of phase velocity is unchanged and equal to $(\sqrt{1+2\gamma}-1) \approx 9.5\%$, due to the same γ in each layer of the model (Table 4). When using different γ in each layer, this ratio will vary for different modes and frequencies (Yuan *et al.* 2022).

We can also find the effects of velocity anisotropy of the VTI medium on Rayleigh waves are relatively more complex than the case of Love waves. The main reason is that the formation of Rayleigh waves is simultaneously related to both *P* and SV waves, whereas the formation of Love waves is solely related to SH waves, and the effects of the VTI medium on the body waves will also be transmitted to the formed surface wave. Thus, Rayleigh waves are controlled by more velocity anisotropy parameters (i.e. ε and δ) than Love waves (i.e. γ). In addition, the effect degree will vary with the change in velocity anisotropy parameters. Ji *et al.* (2020) investigated the effects of different ε and δ on the dispersion characteristic of Rayleigh waves. They found that the effect degree of velocity anisotropy on Rayleigh waves increases markedly with increasing ε (for the given δ), but reduces evidently with increasing δ (for the given ε). The effects of different γ on the dispersion characteristic of Love waves were discussed by Yuan *et al.* (2022). As they found, the increasing γ will significantly increase the effect degree of velocity anisotropy on Love waves.

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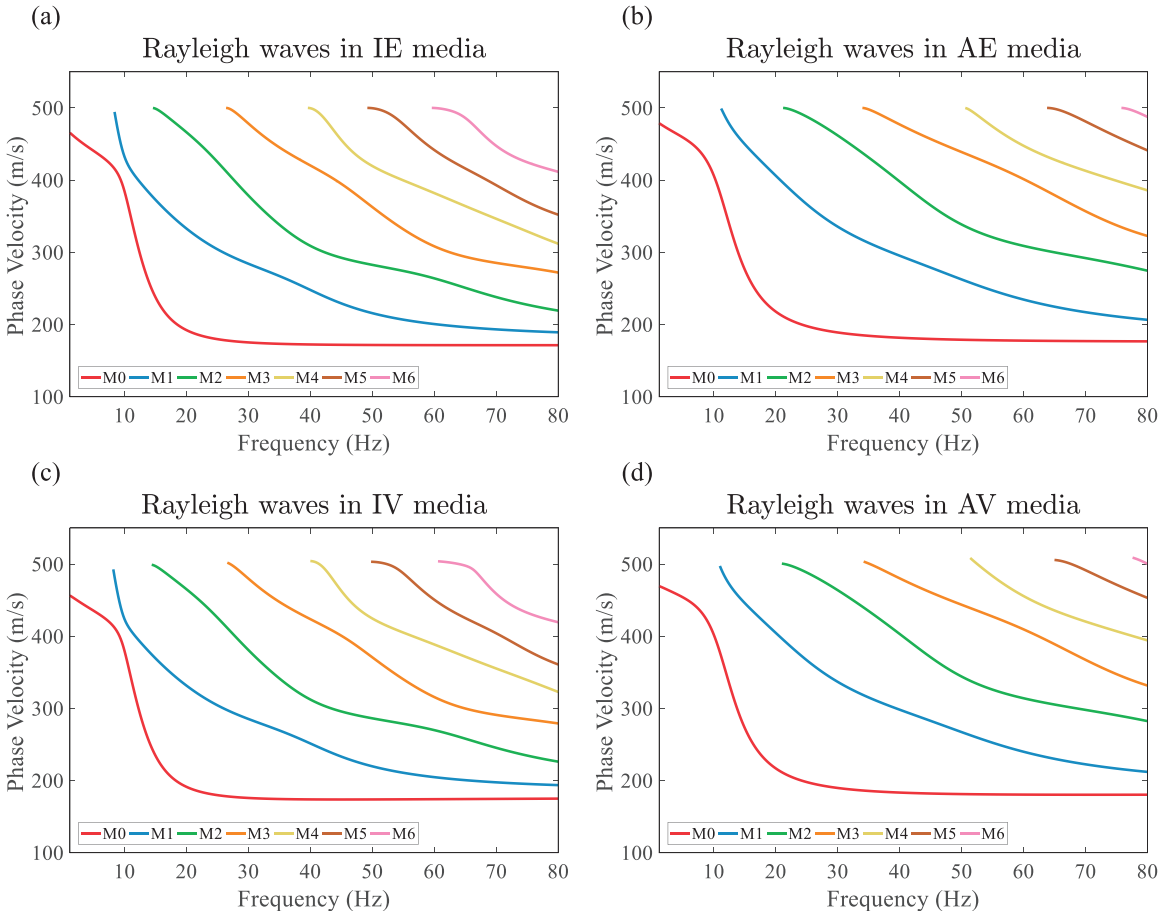


Figure 8. The phase-velocity dispersion curves of Rayleigh waves in (a) isotropic-elastic (IE), (c) isotropic-viscoelastic (IV), (b) anisotropic-elastic (AE) and (d) anisotropic-viscoelastic (AV) media. M0, M1, M2, M3, M4, M5 and M6 represent the fundamental, first-higher, second-higher, third-higher, fourth-higher, fifth-higher and sixth-higher modes, respectively.

4.2 Effects of viscoelasticity

During the propagation of seismic waves within the Earth's interior, the anelasticity of the Earth's media (encompassing the viscoelasticity of solids, the viscosity of pore fluids and melts, and so on) converts the mechanical energy of seismic waves into internal energy, leading to the irreversible dissipation of seismic wave energy (Knopoff 1964). This phenomenon is commonly referred to as intrinsic attenuation (Aki & Richards 1980). Apart from intrinsic attenuation, various factors such as the geometric spreading, focusing/defocusing and scattering (often collectively referred to as scattering attenuation), the excitation and radiation pattern of the seismic source, as well as the instrumental response and site effects during wave propagation can all influence the amplitude strength of seismic waves (e.g. Dai *et al.* 2022). Scattering attenuation can still exist in completely elastic media, and its nature is the redistribution of mechanical energy of seismic waves in time and space, due to the structure heterogeneities of the Earth's media (e.g. Bouchaala *et al.* 2019).

In this study, we only consider the intrinsic attenuation caused by the viscoelasticity of solid media. The effects of viscoelasticity on Rayleigh waves or Love waves are similar in terms of the characteristics of phase velocity dispersion and amplitude attenuation. In elastic layered media, surface wave modal solutions are real wavenumbers (i.e. $k = \omega/c_r$). Therefore, we can only obtain the phase-velocity (c_r) dispersion curves of surface waves, which are well-known as the geometric dispersion resulting from the geometric structure of the velocity profile. This also implies that scattering attenuation will not occur in the surface wave modal solutions. However, when existing viscoelasticity in media, surface wave modal solutions become complex wavenumbers (i.e. $k = \omega/c_r - i\alpha$), and surface waves will not only exhibit additional dispersion features but also carry the intrinsic attenuation (Aki & Richards 1980), due to the conversion of mechanical energy into internal energy. Rix *et al.* (2000) also called them material dispersion and material attenuation, respectively. The intrinsic attenuation of surface waves can be described by their attenuation coefficient (α) curves.

As Figs 7(c), (d), 9(c) and (d) show, in both IV and AV media, both Rayleigh waves and Love waves possess multimodal attenuation coefficient curves including fundamental mode (M0) and higher modes (M1–M6). The attenuation coefficient curves of different modes exhibit different features, but they present an overall trend of increasing attenuation from the low-frequency end to the high-frequency end. It is consistent with the attenuation characteristic of viscoelastic medium, that is, higher-frequency components generally experience more severe attenuation than lower-frequency components. To elaborate further, this is intimately connected with the wavelength ($\lambda = c_r/f$) of surface waves, and typically, there exists a positive correlation between the detection depth of surface waves and their wavelength. In other words, surface waves with shorter wavelengths are generally more sensitive to the shallower

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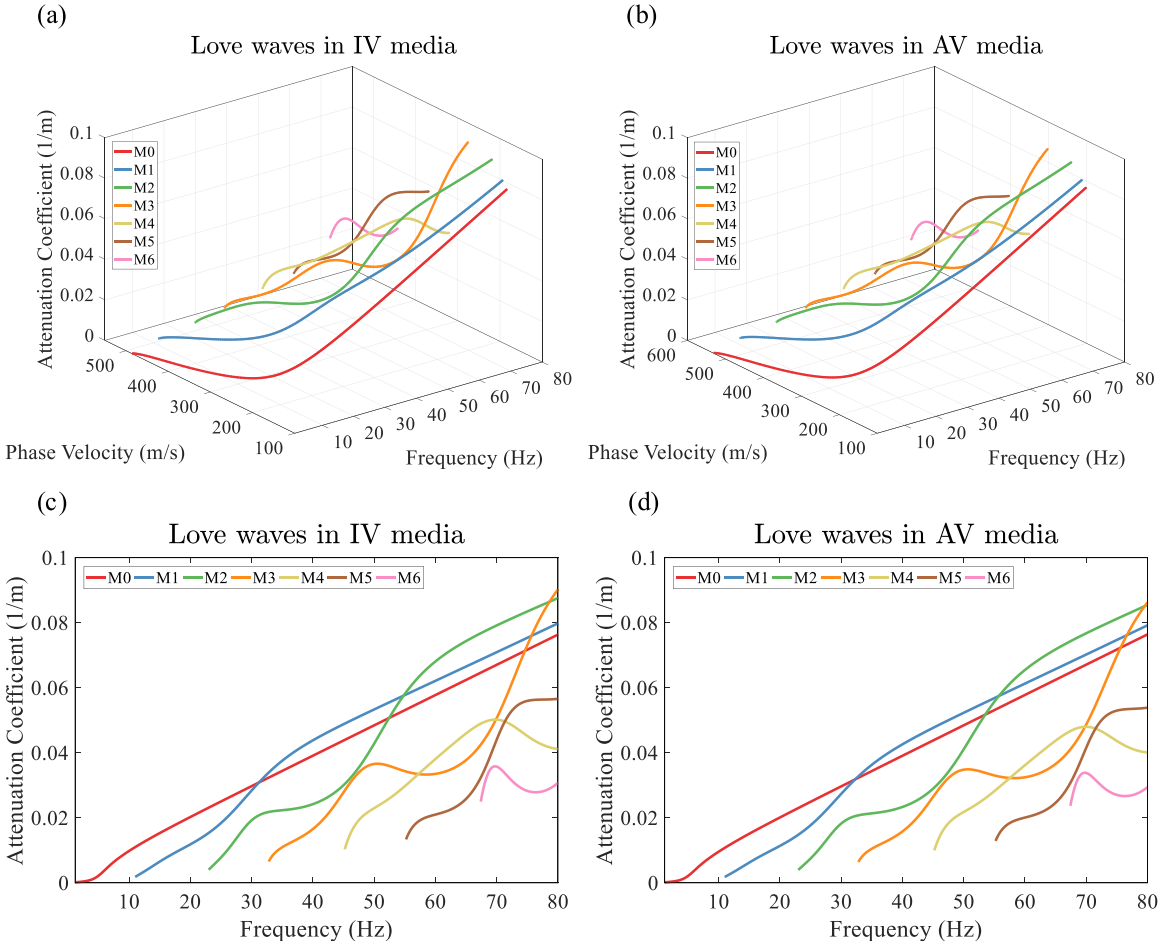


Figure 9. The Love-wave modal solutions in (a, c) isotropic-viscoelastic (IV) and (b, d) anisotropic-viscoelastic (AV) media. (a) and (b) correspond to the results in the frequency-phase velocity-attenuation coefficient domain; (c) and (d) correspond to their projections to the frequency-attenuation coefficient domain, that is the attenuation coefficient curves. M0, M1, M2, M3, M4, M5 and M6 represent the fundamental, first-higher, second-higher, third-higher, fourth-higher, fifth-higher and sixth-higher modes, respectively.

strata, while the shallower strata possess lower S-wave quality factors (i.e. Q_{S0} values) which means stronger attenuation. In addition, the attenuation coefficient curves of different modes seem to intersect with each other, but they are actually separated from each other in the 3-D space of the frequency-phase velocity-attenuation coefficient domain (Figs 7a, b, 9a and b).

The material dispersion of surface waves can be analysed by comparing the results in IV and IE media. As Figs 8(a), (c), 10(a) and (c) show, although the dispersion curve shapes of both Rayleigh waves and Love waves are basically unchanged in both IV and IE media, it is recognizable that the phase velocity values in IV media are slightly greater than those in IE media in the high-frequency end. For example, at the frequency of 80 Hz, the phase velocities of the fundamental mode (M0) and sixth-higher mode (M6) of Rayleigh waves in IV media are about 174.92 and 419.36 m s^{-1}, respectively, while those in IE media are about 171.37 and 411.51 m s^{-1}. As shown in Figs 11(a) and (b), these weak differences can be reflected better by the phase velocity differences of dispersion curves in IV media to those in IE media. Whether for Rayleigh waves or Love waves, the overall trend of the phase velocity differences increases with the increase of frequency, and the maximum absolute difference is not more than 17 m s^{-1} (while the maximum increasing ratio is not more than 4 per cent). In viscoelastic media, the phase velocities and attenuation coefficients of surface waves are coupled in the form of complex wavenumbers. The material dispersion of surface waves can be attributed to and accompanied by their intrinsic attenuation. In addition, since the reference frequency f_r is set to 25 Hz in the viscoelasticity mechanism (eq. 3), the phase velocity differences are almost equal to zero at 25 Hz (especially for fundamental mode), which means that the phase velocities in IV and IE media are almost equal at this frequency (see black dashed lines). When $f > f_r$, the phase velocities in IV media are greater than those in IE media, but the results are opposite when $f < f_r$.

Similarly, the material dispersion of surface waves can also be revealed by comparing the results in AV and AE media (i.e. comparing Figs 11e and f with Figs 11c and d). Moreover, according to the works of Yuan *et al.* (2018, 2022), the effects of viscoelasticity on surface waves, namely material dispersion and intrinsic attenuation, are controlled by the quality factor (see eqs 3 and 8) and will increase with decreasing quality factor.

4.3 Effects of attenuation anisotropy

The attenuation anisotropy of viscoelastic-TI media is described by three dimensionless Thomsen-style parameters (i.e. ε_Q, δ_Q and γ_Q). Parameters ε_Q and δ_Q are aimed at the case of Rayleigh waves,

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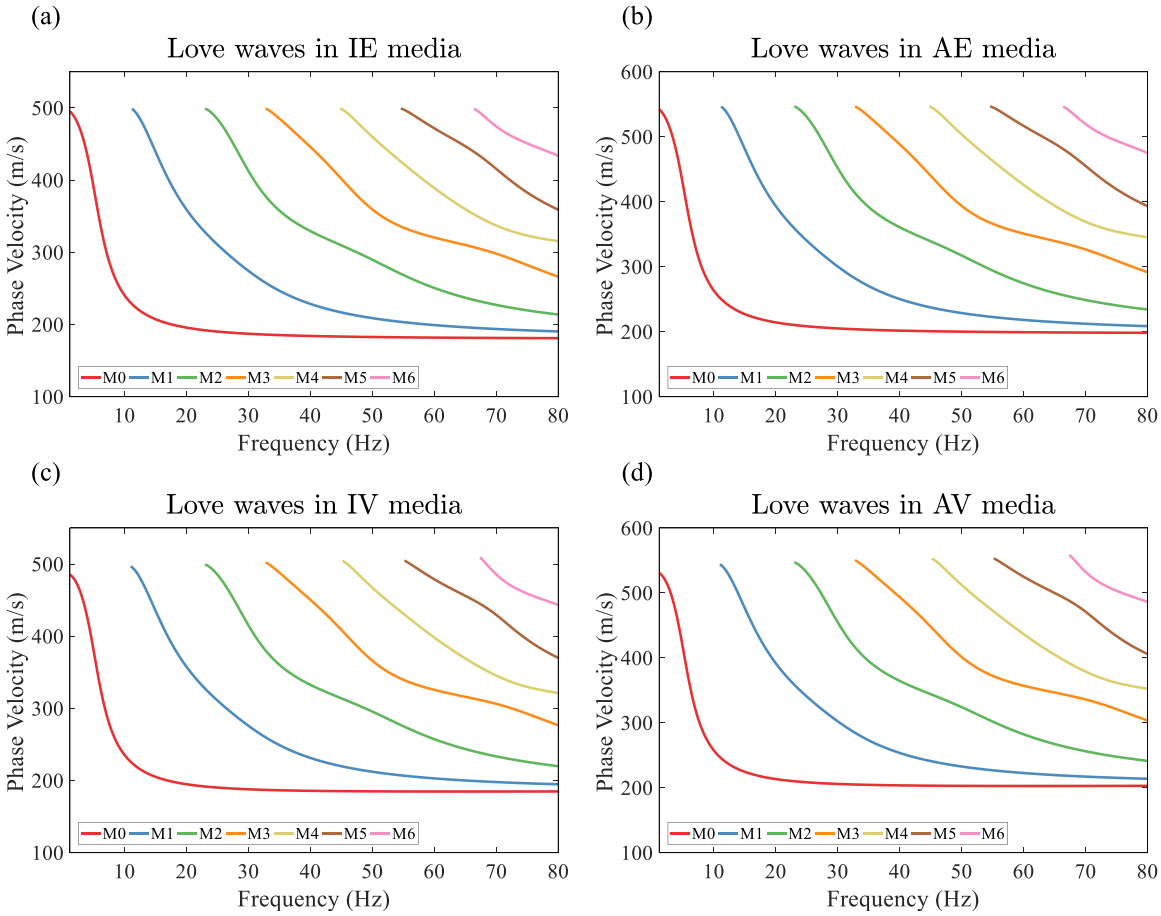


Figure 10. The phase-velocity dispersion curves of Love waves in (a) isotropic-elastic (IE), (c) isotropic-viscoelastic (IV), (b) anisotropic-elastic (AE) and (d) anisotropic-viscoelastic (AV) media. M0, M1, M2, M3, M4, M5 and M6 represent the fundamental, first-higher, second-higher, third-higher, fourth-higher, fifth-higher and sixth-higher modes, respectively.

while parameter γ_Q is used for the case of Love waves. The attenuation anisotropy means that the viscoelasticity is direction-dependent, as the quality factors of body waves in Fig. 2 show. By comparing with the modal solutions in attenuation isotropy media, we can analyse the effects of attenuation anisotropy on the surface-wave material dispersion and intrinsic attenuation. The results in AV media with attenuation isotropy can be simulated by setting the attenuation anisotropy parameter (i.e. ε_Q, δ_Q and γ_Q) to zero. The material dispersion of surface waves is indicated by the phase velocity differences of dispersion curves in AV media to those in AE media, while the attenuation coefficient curves of surface waves in AV media denote their intrinsic attenuation.

The first row of Fig. 12 shows the comparisons of phase velocity differences and attenuation coefficient curves of Rayleigh waves between attenuation anisotropy with different ε_Q (i.e. setting ε_Q to 0.1 and 0.15, respectively, and fixing δ_Q to 0.05) and attenuation isotropy (i.e. setting both δ_Q and ε_Q to 0.0), while the second row of Fig. 12 displays those between attenuation anisotropy with different δ_Q (i.e. setting δ_Q to 0.05 and 0.1, respectively, and fixing ε_Q to 0.1) and attenuation isotropy. The comparisons of phase velocity differences and attenuation coefficient curves of Love waves between attenuation anisotropy with different γ_Q (i.e. setting γ_Q to 0.1 and 0.2, respectively) and attenuation isotropy (i.e. setting γ_Q to 0.0) are shown in the last row of Fig. 12. Except for the parameters to be discussed, all other parameters in Tables 3 and 4 remain unchanged. For a clearer presentation, we only show the results of the fundamental (M0), second-higher (M2), fourth-higher (M4) and sixth-higher (M6) modes.

As Fig. 12 shows, under the conditions of all ε_Q, δ_Q and γ_Q greater than zero as well as $\varepsilon_Q \geq \delta_Q$, all results with attenuation anisotropy (blue lines) present stronger material dispersion and intrinsic attenuation than those with attenuation isotropy (red lines). Similar to the case of velocity anisotropy, the effect degree of attenuation anisotropy will vary with the change in attenuation anisotropy parameters. When increasing ε_Q from 0.1 (light blue dashed–dotted lines) to 0.15 (dark blue dashed line), the effect degree on Rayleigh waves increases markedly (Figs 12a and b), but reduces evidently when increasing δ_Q from 0.05 to 0.1 (Figs 12c and d). As shown in Figs 12(e) and (f), the effect degree of attenuation anisotropy on Love waves will significantly increase when γ_Q increases from 0.1 (light blue dashed–dotted lines) to 0.2 (dark blue dashed line). In addition, we can also notice that the larger difference between ε_Q and δ_Q will result in a greater effect degree of attenuation anisotropy on Rayleigh waves.

4.4 Eigenfunctions and ellipticity

After obtaining the modal solutions of surface waves, we can further determine the corresponding eigenfunctions including eigendisplacements and eigentractions by following Appendix D. Using

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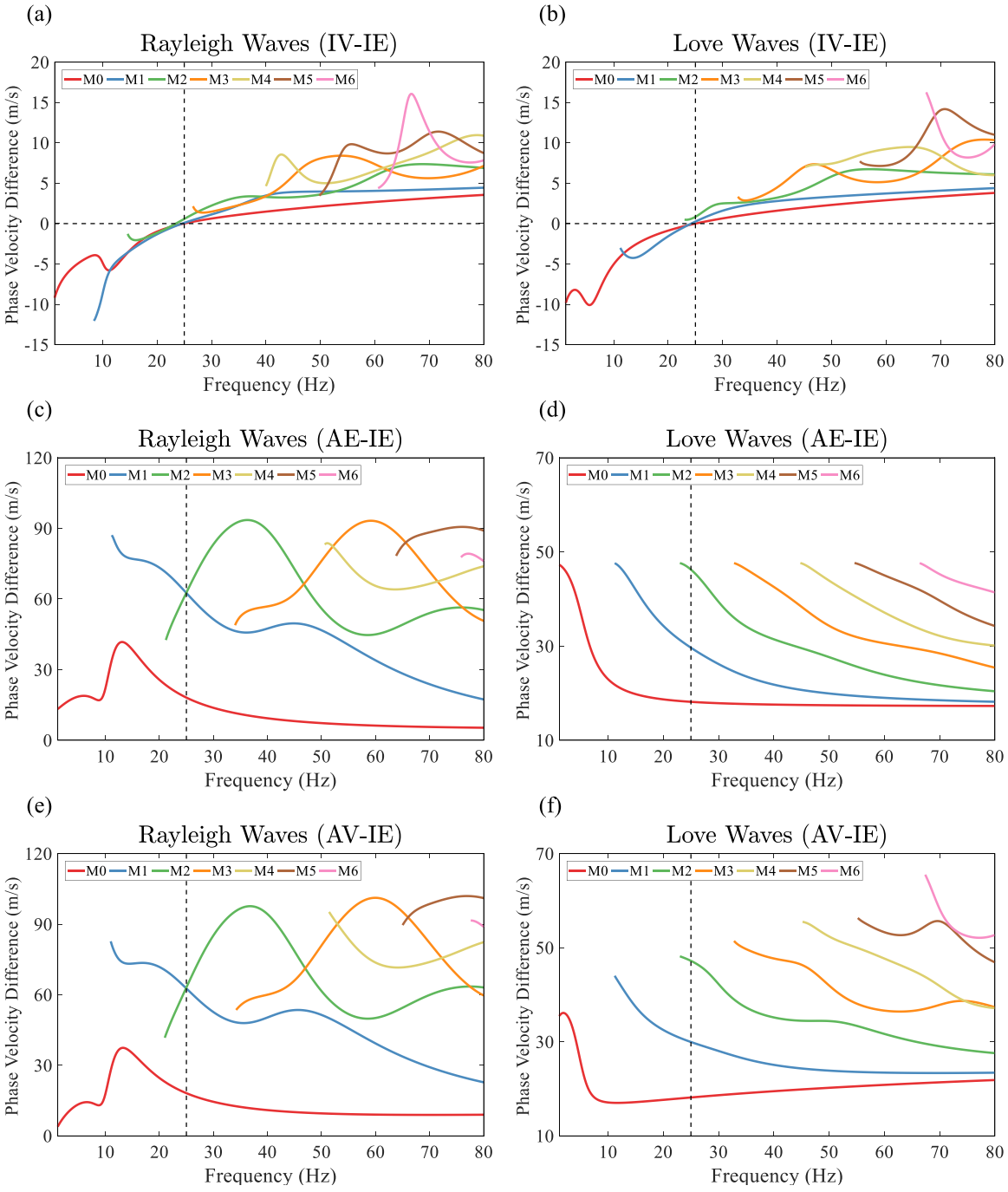


Figure 11. The phase-velocity differences of dispersion curves of Rayleigh waves (a, c and e) and Love waves (b, d and f) in IV, AE and AV media to those in IE media. The black dashed lines in vertical and horizontal directions represent the frequency of 25 Hz and the phase velocity difference of 0, respectively.

models in Tables 3 and 4, we calculate the eigendisplacements and eigentractions of Rayleigh waves and Love waves in IE, IV, AE and AV media at the frequency of 80 Hz and show them in Fig. 13. In fact, for any given eigenvalue (i.e. wavenumber k), the corresponding eigenfunctions are not unique. In other words, the eigenfunctions obtained from each calculation are only specific solutions, and the results after multiplying them by any non-zero constant are still the eigenfunctions of this eigenvalue. Thus, the amplitudes of eigenfunctions for different modes as well as in different media have significant differences, which means that comparing their amplitudes directly is meaningless. For a better presentation, each trace in Fig. 13 is normalized relative to its own maximum value in the depth range. This normalized processing of eigenfunctions has also been used in previous works (i.e. Chen 1993; Wu & Chen 2016). Fig. 13 can reveal the changes in eigendisplacements and eigentractions of different modes in different media with depth.

As Fig. 13 shows, the polarity of eigenfunctions usually changes alternately with increasing depth and their amplitude finally approaches zero at a certain depth. At the given frequency (i.e. 80 Hz), the penetration depth of surface waves increases with increasing mode order (i.e. in order from M0 to M6), which also means that

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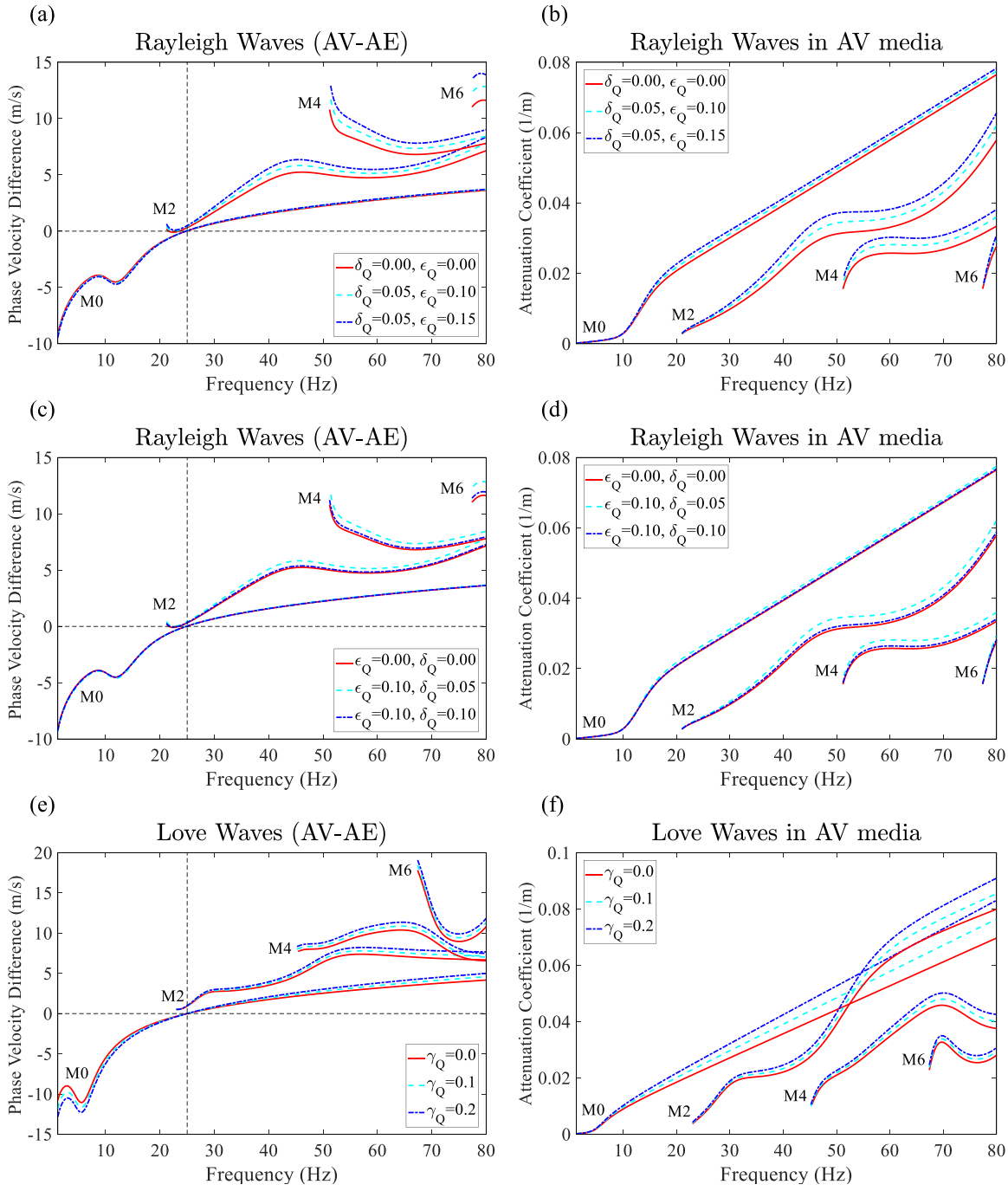


Figure 12. The comparisons of phase velocity differences (left-hand panel) and attenuation coefficient curves (right-hand panel) of surface waves between attenuation anisotropy and attenuation isotropy (red lines): (a) and (b) for Rayleigh waves with different ε_Q; (c) and (d) for Rayleigh waves with different δ_Q; (e) and (f) for Love waves with different γ_Q. The phase velocity differences are calculated by subtracting the dispersion curves in AE media from those in AV media. M0, M2, M4 and M6 represent the fundamental, second-higher, fourth-higher and sixth-higher modes, respectively. The black dashed lines in vertical and horizontal directions represent the frequency of 25 Hz and the phase velocity difference of 0, respectively.

higher modes possess greater sensitivity to deep layers than the fundamental mode. At the depths of some layer interfaces, remarkable changes in eigenfunctions can be identified (e.g. Figs 13b and f). At the free surface, eigendisplacements are non-zero, but eigentractions are theoretically zero due to the boundary condition of the free surface. Thus, the values of eigentractions at the free surface reflect the accuracy of numerical calculations. For example, for Love waves in IE media, the corresponding values from M0 to M6 are about −2.65e-9, 9.17e-10, −7.16e-9, 7.73e-9, −4.86e-8, 1.71e-10 and −5.98e-9, respectively, which demonstrates that the numerical calculations in this study are accurate enough.

The differences in different media can be clearly observed from the eigenfunctions of Rayleigh waves (Figs 13a–d). Except for slight differences in amplitude, the eigenfunctions in IE (red solid lines) and IV (green dashed lines) media present basically consistent shapes, and the case for AE (light blue solid lines) and AV

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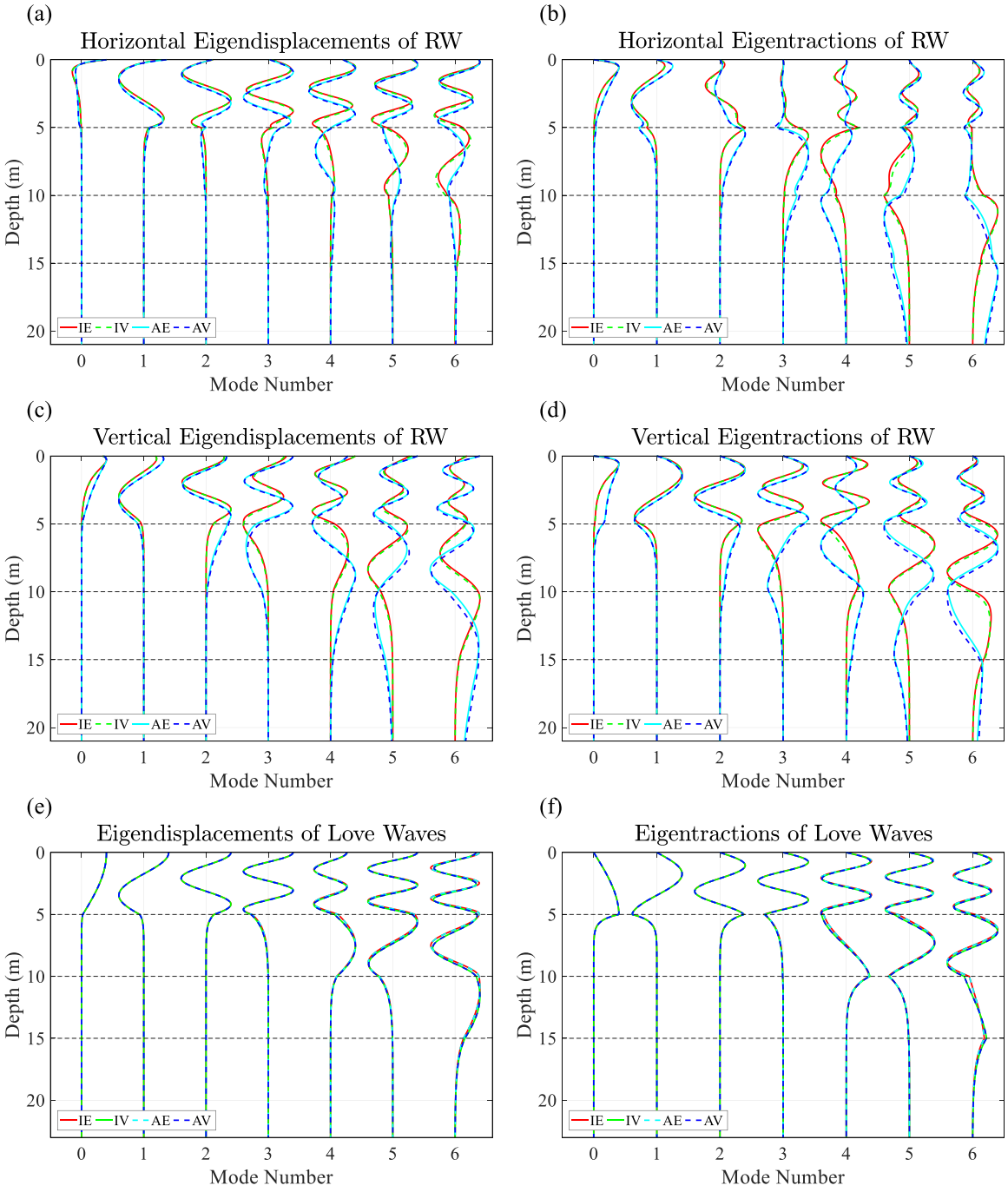


Figure 13. The eigendisplacements (a, c, e) and eigentractions (b, d, f) of surface waves in IE (red solid lines), IV (green dashed lines), AE (light blue solid lines) and AV (dark blue dashed line) media at the frequency of 80 Hz. The first and second rows correspond to Rayleigh waves (i.e. RW), and the last row corresponds to Love waves. The modes from 0 to 6 represent the fundamental, first-higher, second-higher, third-higher, fourth-higher, fifth-higher and sixth-higher modes in sequence. The three horizontal black dashed lines represent the layer interfaces with depths of 5, 10 and 15 m, respectively. Each trace is normalized relative to its own maximum value in the depth range.

(dark blue dashed lines) media is similar. However, the shapes of eigenfunctions are obviously different between the anisotropic (i.e. AE and AV) and isotropic (i.e. IE and IV) media, and the results in the anisotropic media have greater penetration depth than those in isotropic media. It indicates that the effects of velocity anisotropy on the eigenfunctions of Rayleigh waves are more significant than that of viscoelasticity. Differing from the case of Rayleigh waves, except for slight differences in higher modes (e.g. M4, M5 and M6), the eigenfunctions of Love waves in different media are substantially consistent, and these slight but identifiable differences exist between the viscoelastic (i.e. IV and AV) and elastic (i.e. IE and AE) media, which explains the effects of viscoelasticity on the eigenfunctions of Love waves are greater than that of velocity anisotropy (Figs 13e and f).

The horizontal-to-vertical spectral ratios (HVSR) curves (i.e. ellipticity) of Rayleigh waves have been widely proven to estimate the

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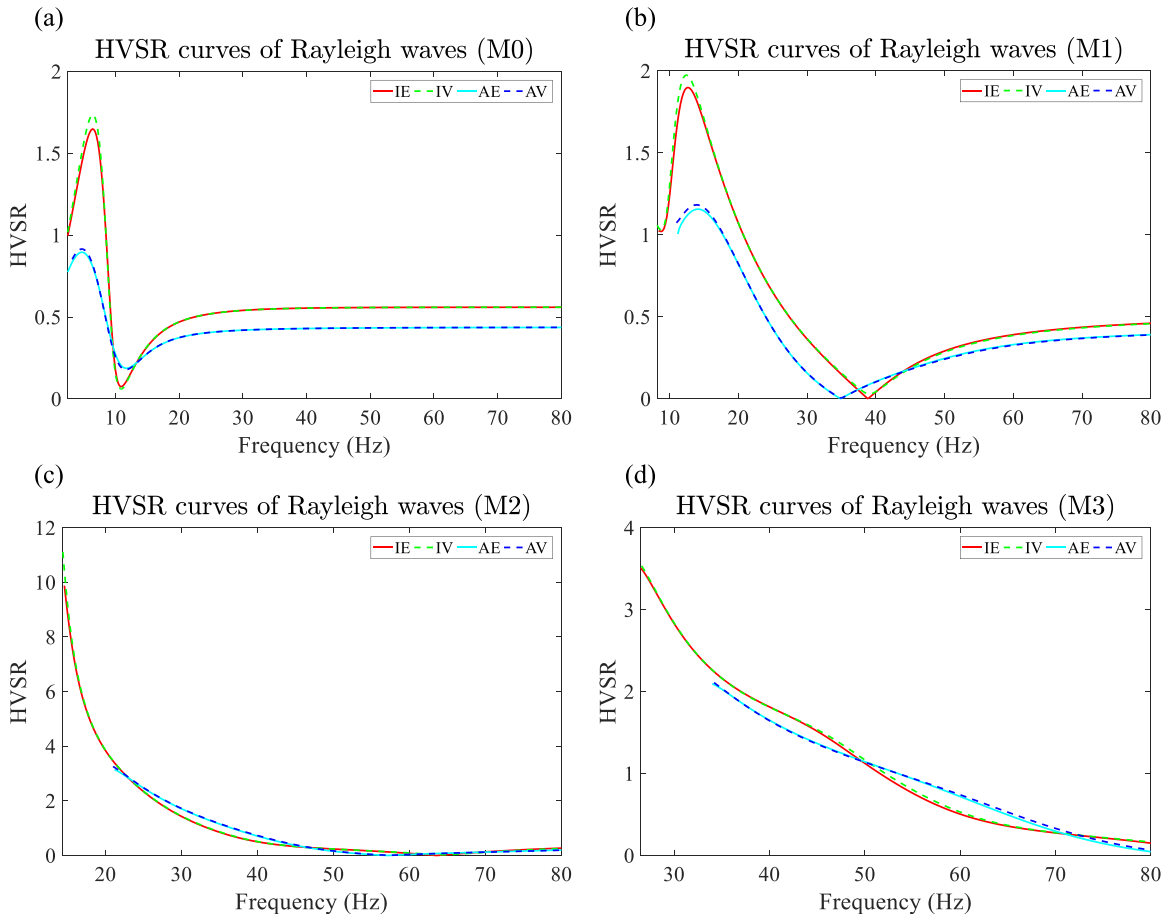


Figure 14. The HVSR curves of Rayleigh waves in IE (red solid lines), IV (green dashed lines), AE (light blue solid lines) and AV (dark blue dashed line) media: (a) fundamental mode (M0); (b) first-higher mode (M1); second-higher mode (M2) and (d) third-higher mode (M3).

near-surface *S*-wave velocity structures by their individual inversion or joint inversion with dispersion curves of Rayleigh waves (e.g. Xu *et al.* 2022). In this study, through the determined horizontal and vertical eigendisplacements, we can further calculate the theoretical multimodal HVSR curves of Rayleigh waves in IE, IV, AE and AV media, as introduced in Appendix E. As Fig. 14 shows, the HVSR curves of different modes present different features. Similar to the eigenfunctions of Rayleigh waves, the shapes of HVSR curves are basically consistent in viscoelastic (e.g. IV, green dashed lines) and elastic (e.g. IE, red solid lines) media, but obviously different between the anisotropic (i.e. AE and AV) and isotropic (i.e. IE and IV) media. It indicates that the effects of velocity anisotropy on the HVSR curves of Rayleigh waves are more prominent than that of viscoelasticity.

5 CONCLUSIONS

This work proposes a systematic and effective method, which is an extended version of the GRTM (Chen 1993; Wu & Chen 2016), for calculating the phase-velocity (c_r) dispersion curves, attenuation-coefficient (α) curves, and eigenfunctions of both Rayleigh and Love waves as well as the ellipticity of Rayleigh waves in layered viscoelastic-VTI media. Under the theory of viscoelastic-VTI media, we start from the constitutive relationship to deduce the source-free dynamic equation (i.e. governing equation) and finally derive a family of secular functions by following the latest framework of GRTM. The entire derivation process is systematically given with explicit expressions. We summarize the step-by-step procedures of implementing our algorithm, where the numerical scheme of combining the root-searching method with the local optimization method is built for determining the complex-valued modal solutions (i.e. complex wavenumber $k = \omega/c_r - i\alpha$) of surface waves. In six-layer IE, IV, AE and AV media, we compare the modal solutions from our GRTM with those from other methods (i.e. the Thomson–Haskell method and the reduced delta matrix method), respectively. The excellent comparisons confirm the correctness of our algorithm.

Based on the developed algorithm, we further compute the modal solutions of both Rayleigh and Love waves in four-layer IE, IV, AE and AV media, and then explore the effects of velocity anisotropy, viscoelasticity and attenuation anisotropy on the dispersion and attenuation characteristics of surface waves in detail by comparing the results of different media. We find that the velocity anisotropy of the medium can significantly change the phase-velocity magnitude and dispersion-curve distribution range of both Rayleigh and Love waves, but the effects on Rayleigh waves are different from the case of Love waves and also relatively more complex. Due to the viscoelasticity of the medium, surface waves will not only exhibit additional phase-velocity dispersion features (called material dispersion) but also possess multimodal attenuation coefficient curves (known as intrinsic attenuation), and the effects are similar whether on Rayleigh waves or Love waves. The attenuation anisotropy of the

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medium indicates that the viscoelasticity degree of the medium is direction-dependent, and the effect degree varies with the change in attenuation anisotropy parameters. In addition, we also discuss the differences in IE, IV, AE and AV media for the eigendisplacements and eigentractions of both Rayleigh and Love waves as well as the ellipticity of Rayleigh waves.

In this study, for the anisotropy of the medium, we only focus on VTI medium, which is the simplest anisotropy type of TI medium and mainly applicable to studying the radial anisotropy of shear wave velocity using Rayleigh wave and Love wave dispersion curves. Actually, besides VTI medium, the TI medium also includes horizontal transversely isotropic (HTI) medium and tilted transversely isotropic (TTI) medium. It will be worthwhile work in our future to investigate how to calculate the dispersion curve of surface waves by GRTM in layered TTI or HTI media (where *P*-SV waves and SH waves may be coupled), and how to study the azimuthal anisotropy of shear wave velocity using surface wave dispersion within the framework of the HTI medium. Furthermore, for the intrinsic attenuation, we only consider the viscoelasticity of solid media in this study. For geological environments that have well-developed pores and are rich in fluids, studying and applying the effects of pore fluids in the media on surface wave dispersion and attenuation characteristics could be valuable work for the detection and monitoring of groundwater, oil and gas reservoirs. In addition, through this work, we have a clear understanding of the dispersion and attenuation characteristics of surface waves in layered VTI and/or viscoelastic media. Consequently, considering the anisotropy and/or viscoelasticity of the medium to conduct the sensitivity analysis in theory and then to establish the surface wave inversion are also worthwhile works in our future, which have the potential to better investigate the solid Earth's internal structure. Our developed algorithm in this study can provide a theoretical basis and useful tool for this purpose.

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SUPPORTING INFORMATION

Supplementary data are available at *GJIRAS* online.

Figure S1. Comparisons of Love-wave modal solutions between our GRTM (red solid lines) and the Thomson–Haskell method (labelled as THM; blue points) for Model 1 in (a) isotropic-elastic (IE), (b) isotropic-viscoelastic (IV), (c) anisotropic-elastic (AE) and (d) anisotropic-viscoelastic (AV) media.

Figure S2. Comparisons of Love-wave modal solutions between our GRTM (red solid lines) and the Thomson–Haskell method (labelled as THM; blue points) for Model 2 in (a) isotropic-elastic (IE), (b) isotropic-viscoelastic (IV), (c) anisotropic-elastic (AE) and (d) anisotropic-viscoelastic (AV) media.

Figure S3. Comparisons of phase-velocity dispersion curves of Love waves for Model 2: (a, b) comparing the accurate solutions (red solid lines) with approximate results (blue dashed–dotted lines) in isotropic-viscoelastic (IV) and anisotropic-viscoelastic (AV) media; (c, d) comparing the accurate results in IV and AV media (red solid lines) with those in IE and AE media (blue dashed–dotted lines).

Table S1. Parameters of Model 1 used for the SH-wave case.

Table S2. Parameters of Model 2 used for the SH-wave case.

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CONFLICT OF INTEREST

All authors declare that they have no conflicts of interest.

DATA AVAILABILITY

The code and data used in this study will be shared upon reasonable request to the corresponding author.

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APPENDIX A: VELOCITY AND ATTENUATION ANISOTROPIES OF BODY WAVES

The free dynamic equation of viscoelastic-VTI medium in the frequency domain is given in eq. (B3). When considering the general plane-wave solutions of the displacement vector, the differential operator matrices ∇_1 and ∇_2 in eq. (B3) can be replaced with the following forms in the Cartesian coordinate system (Carcione 2015):

$$\begin{cases} \nabla_1 \rightarrow -ik \begin{pmatrix} l_x & 0 & 0 & 0 & l_z & l_y \\ 0 & l_y & 0 & l_z & 0 & l_x \\ 0 & 0 & l_z & l_y & l_x & 0 \end{pmatrix} \equiv -ik\mathbf{L} \\ \nabla_2 = \nabla_1^{\mathrm{T}} = -ik\mathbf{L}^{\mathrm{T}} \end{cases}, \tag{A1}$$

where k is the size of the wavenumber vector and is complex in a viscoelastic medium; l_x, l_y and l_z represent the direction cosines of the wavenumber vector. Consequently, we obtain the Kelvin–Christoffel equation:

$$\left(k^2\mathbf{\Gamma} - \rho\omega^2\mathbf{I}\right) \cdot \tilde{\mathbf{u}}(\mathbf{x}, \omega) = 0, \tag{A2}$$

where $\mathbf{\Gamma} = \mathbf{L} \cdot \mathbf{M}(\mathbf{x}, \omega) \cdot \mathbf{L}^{\mathrm{T}}$ is the Kelvin–Christoffel matrix with 3×3 form; $\mathbf{I}$ is an identity matrix. Eq. (A2) is a characteristic equation, and through solving the following dispersion relation:

$$\det\left(k^2\mathbf{\Gamma} - \rho\omega^2\mathbf{I}\right) = 0, \tag{A3}$$

we can obtain three characteristic wavenumbers k_ξ ($\xi = qP, qSV, SH$), which correspond to three mode types of body waves [i.e. quasi-P (qP), quasi-SV (qSV) and SH waves] propagating in the homogeneous full-space medium. For any given frequency, $k_\xi(l_x, l_y, l_z)$ defines a surface with directional cosines as a function.

Taking the 2-D case in the $x-z$ plane as an example ($l_y = 0$), the complex velocities of quasi-P, quasi-SV and SH waves (using ω/k_m) can be defined by:

$$V_{SH}(\omega, \theta) = \sqrt{\frac{M_{66}(\omega)l_x^2(\theta) + M_{55}(\omega)l_z^2(\theta)}{\rho}}, \tag{A4}$$

$$V_{qSV}(\omega, \theta) = \sqrt{\frac{M_{11}(\omega)l_x^2(\theta) + M_{33}(\omega)l_z^2(\theta) + M_{55}(\omega) - E(\omega, \theta)}{2\rho}}, \tag{A5}$$

$$V_{qP}(\omega, \theta) = \sqrt{\frac{M_{11}(\omega)l_x^2(\theta) + M_{33}(\omega)l_z^2(\theta) + M_{55}(\omega) + E(\omega, \theta)}{2\rho}}, \tag{A6}$$

$$E(\omega, \theta) = \sqrt{\left(\left[M_{11}(\omega) - M_{55}(\omega)\right]l_x^2(\theta) + \left[M_{55}(\omega) - M_{33}(\omega)\right]l_z^2(\theta)\right)^2 + 4\left(\left[M_{13}(\omega) + M_{55}(\omega)\right]l_x(\theta)l_z(\theta)\right)^2}. \tag{A7}$$

When defining the propagation angle of the wavenumber vector direction relative to the symmetry axis (z-axis) of the VTI medium as θ, we have $l_x(\theta) = \sin\theta$ and $l_z(\theta) = \cos\theta$. As eqs (A4)–(A7) show, in a viscoelastic-VTI medium, the P-SV wave modes are coupled together (M_{11} and M_{33} corresponding to P waves, while M_{55} corresponding to S waves). Thus, they are known as the quasi-P, quasi-SV wave modes. The SH-wave mode, which is decoupled with the P-SV wave modes, is a purely shear mode (both M_{55} and M_{66} corresponding to S waves). The vertical and horizontal directions represent the purely modal direction, where the P-wave polarization is consistent with the direction of the wave vector, and the S-wave polarization is perpendicular to the direction of the wave vector. In this case, all three wave modes are decoupled, as follows:

$$V_{SH}(\omega, 0^\circ) = \sqrt{M_{55}(\omega)/\rho}, \quad V_{SV}(\omega, 0^\circ) = \sqrt{M_{55}(\omega)/\rho},$$

$$V_P(\omega, 0^\circ) = \sqrt{M_{33}(\omega)/\rho}, \tag{A8}$$

$$V_{SH}(\omega, 90^\circ) = \sqrt{M_{66}(\omega)/\rho}\ , \ V_{SV}(\omega, 90^\circ) = \sqrt{M_{55}(\omega)/\rho}\ ,$$

$$V_P(\omega, 90^\circ) = \sqrt{M_{11}(\omega)/\rho}. \tag{A9}$$

Corresponding to the complex velocities, the quality factors, phase velocities and attenuation coefficients of these three wave modes can be expressed as:

$$Q_\xi(\omega, \theta) = \frac{\mathrm{Re}\left[V_\xi^2(\omega, \theta)\right]}{\mathrm{Im}\left[V_\xi^2(\omega, \theta)\right]}, \tag{A10}$$

$$V_{p,\xi}(\omega, \theta) = 1/\mathrm{Re}\left[1/V_\xi(\omega, \theta)\right], \tag{A11}$$

$$\alpha_\xi(\omega, \theta) = -\mathrm{Im}\left[\omega/V_\xi(\omega, \theta)\right], \tag{A12}$$

where $\xi = qP, qSV, SH$; the subscript letter p represents the English word phase. Therefore, the quality factors, phase velocities and attenuation coefficients of these three wave modes are frequency and direction-dependent.

APPENDIX B: GOVERNING EQUATIONS AND GENERAL SOLUTIONS

By combining the constitutive equation (eq. 1) with the following momentum conservation equation (without source term) and geometric equation (Lai *et al.* 2010):

$$-\rho^{(j)}\omega^2\mathbf{u}^{(j)}(\mathbf{x}, \omega) = \nabla_1 \cdot \mathbf{T}^{(j)}(\mathbf{x}, \omega), \tag{B1}$$

$$\mathbf{S}^{(j)}(\mathbf{x}, \omega) = \nabla_2 \cdot \mathbf{u}^{(j)}(\mathbf{x}, \omega), \tag{B2}$$

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in the frequency domain, the source-free dynamic equation of vector form for the viscoelastic-VTI multilayered half-space model can be written as follows:

$$-\rho^{(j)}\omega^2\mathbf{u}^{(j)}(\mathbf{x},\omega)=\nabla_1\cdot\left[\mathbf{M}^{(j)}(\mathbf{x},\omega)\left(\nabla_2\cdot\mathbf{u}^{(j)}(\mathbf{x},\omega)\right)\right], \tag{B3}$$

where $\mathbf{u}^{(j)}=[u_r^{(j)},u_\theta^{(j)},u_z^{(j)}]^{\mathrm{T}}$ is the displacement vector; the differential operator matrices ∇_1 and ∇_2 with Auld notation have the following matrix representations:

$$\nabla_1=\begin{bmatrix}\frac{\partial}{\partial r}+\frac{1}{r} & -\frac{1}{r} & 0 & 0 & \frac{\partial}{\partial z} & \frac{1}{r}\frac{\partial}{\partial\theta}\\ 0 & \frac{1}{r}\frac{\partial}{\partial\theta} & 0 & \frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial r}+\frac{2}{r}\\ 0 & 0 & \frac{\partial}{\partial z} & \frac{1}{r}\frac{\partial}{\partial\theta} & \frac{\partial}{\partial r}+\frac{1}{r} & 0\end{bmatrix}, \tag{B4}$$

$$\nabla_2=\begin{bmatrix}\frac{\partial}{\partial r} & \frac{1}{r} & 0 & 0 & \frac{\partial}{\partial z} & \frac{1}{r}\frac{\partial}{\partial\theta}\\ 0 & \frac{1}{r}\frac{\partial}{\partial\theta} & 0 & \frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial r}-\frac{1}{r}\\ 0 & 0 & \frac{\partial}{\partial z} & \frac{1}{r}\frac{\partial}{\partial\theta} & \frac{\partial}{\partial r} & 0\end{bmatrix}^{T}. \tag{B5}$$

The traction vector $\boldsymbol{\tau}^{(j)}=[\sigma_{zz}^{(j)},\sigma_{\theta z}^{(j)},\sigma_{rz}^{(j)}]^{\mathrm{T}}$ acting at a layer interface with the normal in the z-direction can be given by:

$$\boldsymbol{\tau}^{(j)}(\mathbf{x},\omega)=\mathbf{T}^{(j)}(\mathbf{x},\omega)\cdot\mathbf{e_z}=\left[\mathbf{M}^{(j)}(\mathbf{x},\omega)\left(\nabla_2\cdot\mathbf{u}^{(j)}(\mathbf{x},\omega)\right)\right]\cdot\mathbf{e_z}, \tag{B6}$$

where $\mathbf{e_z}$ represents the vertical unit vector.

The governing equations used in this study consist of eqs (B3) and (B6). The vector functions (i.e. $\mathbf{u}^{(j)}$ and $\boldsymbol{\tau}^{(j)}$) involved in the governing equations can be expanded by a set of vector wave function bases with orthogonality and completeness (see Text S1), which have been widely adopted in the processing of analytical problems (e.g. Ren *et al.* 2012; Tang & Fang 2021; Yuan *et al.* 2021; Yang *et al.* 2023). Consequently, we can derive two systems of first-order ordinary differential equations corresponding to the two decoupled cases of *P*-SV and SH waves, and they can be written in the following unified form (Aki & Richards 1980; Kennett 1983; Chen 1993):

$$\frac{d}{dz}\mathbf{y}^{(j)}(z)=\mathbf{A}^{(j)}\mathbf{y}^{(j)}(z),\ j=1,2,\ldots,N,N+1, \tag{B7}$$

where $\mathbf{y}^{(j)}(z)$ represents the displacement-stress vector; $\mathbf{A}^{(j)}$ is a coefficient matrix. For the *P*-SV wave case, we have:

$$\mathbf{y}^{(j)}=\begin{bmatrix}u_S^{(j)} & u_R^{(j)} & \tau_S^{(j)} & \tau_R^{(j)}\end{bmatrix}^{\mathrm{T}}, \tag{B8}$$

$$\mathbf{A}^{(j)}=\begin{bmatrix}0 & k & \frac{1}{M_{55}^{(j)}} & 0\\ -\frac{M_{13}^{(j)}k}{M_{33}^{(j)}} & 0 & 0 & \frac{1}{M_{33}^{(j)}}\\ M_{11}^{(j)}k^2-\frac{M_{13}^{(j)}M_{13}^{(j)}k^2}{M_{33}^{(j)}}-\rho^{(j)}\omega^2 & 0 & 0 & \frac{M_{13}^{(j)}k}{M_{33}^{(j)}}\\ 0 & -\rho^{(j)}\omega^2 & -k & 0\end{bmatrix}; \tag{B9}$$

for the SH wave case, we have:

$$\mathbf{y}^{(j)}=\begin{bmatrix}u_T^{(j)} & \tau_T^{(j)}\end{bmatrix}^{\mathrm{T}}, \tag{B10}$$

$$\mathbf{A}^{(j)}=\begin{bmatrix}0 & \frac{1}{M_{55}^{(j)}}\\ M_{66}^{(j)}k^2-\rho^{(j)}\omega^2 & 0\end{bmatrix}, \tag{B11}$$

where k is the horizontal complex wavenumber; $u_T^{(j)}$, $u_S^{(j)}$ and $u_R^{(j)}$ as well as $\tau_T^{(j)}$, $\tau_S^{(j)}$ and $\tau_R^{(j)}$ are the expansion coefficients of displacement vector $\mathbf{u}$ and traction vector $\boldsymbol{\tau}$, respectively.

According to the general theory of ordinary differential equations, the general solution of eq. (B7) can be written following form (Chen 1993, 1999):

$$\mathbf{y}^{(j)}(z)=\mathbf{E}^{(j)}\boldsymbol{\Lambda}^{(j)}(z)\mathbf{C}^{(j)},\ j=1,2,\ldots,N,N+1, \tag{B12}$$

where the wave amplitude vector $\mathbf{C}^{(j)}$ is an unknown constant vector to be determined; the square matrix $\mathbf{E}^{(j)}$ and the diagonal matrix $\boldsymbol{\Lambda}^{(j)}$ are two known matrices related to the eigenvectors and eigenvalues of the matrix $\mathbf{A}^{(j)}$, respectively. We derive the explicit expressions of $\mathbf{E}^{(j)}$and $\boldsymbol{\Lambda}^{(j)}$ for both cases of *P*-SV and SH waves, which are given in Appendix C.

APPENDIX C: EXPRESSIONS OF MATRICES Λ AND E

For the *P*-SV wave case, the matrix $\mathbf{A}^{(j)}$ can be written as:

$$\mathbf{A}^{(j)}=\begin{bmatrix}0 & a_1 & a_2 & 0\\ a_3 & 0 & 0 & a_4\\ a_5 & 0 & 0 & -a_3\\ 0 & a_6 & -a_1 & 0\end{bmatrix} \tag{C1}$$

with

$$a_1=k,\ a_2=\frac{1}{M_{55}^{(j)}},\ a_3=-\frac{M_{13}^{(j)}k}{M_{33}^{(j)}},\ a_4=\frac{1}{M_{33}^{(j)}},$$

$$a_5=M_{11}^{(j)}k^2-\frac{M_{13}^{(j)}M_{13}^{(j)}k^2}{M_{33}^{(j)}}-\rho^{(j)}\omega^2,\ a_6=-\rho^{(j)}\omega^2. \tag{C2}$$

$\mathbf{E}^{(j)}$ and $\boldsymbol{\Lambda}^{(j)}$ are two matrices related to the eigendecomposition of the matrix $\mathbf{A}^{(j)}$. Matrix $\boldsymbol{\Lambda}^{(j)}$ has the form:

$$\boldsymbol{\Lambda}^{(j)}(z)=\begin{bmatrix}\boldsymbol{\Lambda}_d^{(j)}(z) & \mathbf{0}\\ \mathbf{0} & \boldsymbol{\Lambda}_u^{(j)}(z)\end{bmatrix}, \tag{C3}$$

$$\boldsymbol{\Lambda}_d^{(j)}(z)=\mathrm{diag}\left\{\exp\left[-\lambda_P^{(j)}\left(z-z^{(j-1)}\right)\right]\quad\exp\left[-\lambda_{SV}^{(j)}\left(z-z^{(j-1)}\right)\right]\right\}, \tag{C4}$$

$$\boldsymbol{\Lambda}_u^{(j)}(z)=\mathrm{diag}\left\{\exp\left[-\lambda_P^{(j)}\left(z^{(j)}-z\right)\right]\quad\exp\left[-\lambda_{SV}^{(j)}\left(z^{(j)}-z\right)\right]\right\}, \tag{C5}$$

where the subscripts d, u, P and SV denote the downgoing waves, upgoing waves, *P* waves and SV waves, respectively; $-\lambda_P^{(j)}$, $-\lambda_{SV}^{(j)}$, $\lambda_P^{(j)}$ and $\lambda_{SV}^{(j)}$ are four eigenvalues of the matrix $\mathbf{A}^{(j)}$. $\lambda_P^{(j)}$ and $\lambda_{SV}^{(j)}$ can be calculated by:

$$\lambda_P^{(j)}=\sqrt{\frac{t_2+\sqrt{t_1}}{2}},\ \mathrm{Re}\left[\lambda_P^{(j)}\right]>0, \tag{C6}$$

$$\lambda_{SV}^{(j)}=\sqrt{\frac{t_2-\sqrt{t_1}}{2}},\ \mathrm{Re}\left[\lambda_{SV}^{(j)}\right]>0 \tag{C7}$$

with

$$t_1=4a_1a_2a_3a_5+4a_1a_3a_4a_6+a_2^2a_5^2+a_4^2a_6^2-4a_1^2a_4a_5-4a_2a_3^2a_6-2a_2a_4a_5a_6, \tag{C8}$$

$$t_2=2a_1a_3+a_2a_5+a_4a_6. \tag{C9}$$

The matrix $\mathbf{E}^{(j)}$ can be expressed as:

$$\mathbf{E}^{(j)}=\begin{bmatrix}\mathbf{E}_{d,P}^{(j)} & \mathbf{E}_{d,SV}^{(j)} & \mathbf{E}_{u,P}^{(j)} & \mathbf{E}_{u,SV}^{(j)}\end{bmatrix}, \tag{C10}$$

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$$\mathbf{E}_{d,P}^{(j)} = \begin{bmatrix} \frac{a_1}{a_6} - \frac{a_1^2+a_2a_6}{a_6} f(-\lambda_P^{(j)}) \\ -\lambda_P^{(j)} \left(\frac{1}{a_6} - \frac{a_1}{a_6} f(-\lambda_P^{(j)}) \right) \\ \lambda_P^{(j)} f(-\lambda_P^{(j)}) \\ 1 \end{bmatrix},$$

$$\mathbf{E}_{d,SV}^{(j)} = \begin{bmatrix} \frac{a_1}{a_6} - \frac{a_1^2+a_2a_6}{a_6} f(-\lambda_{SV}^{(j)}) \\ -\lambda_{SV}^{(j)} \left(\frac{1}{a_6} - \frac{a_1}{a_6} f(-\lambda_{SV}^{(j)}) \right) \\ \lambda_{SV}^{(j)} f(-\lambda_{SV}^{(j)}) \\ 1 \end{bmatrix}, \tag{C11}$$

$$\mathbf{E}_{u,P}^{(j)} = \begin{bmatrix} \frac{a_1}{a_6} - \frac{a_1^2+a_2a_6}{a_6} f(\lambda_P^{(j)}) \\ \lambda_P^{(j)} \left(\frac{1}{a_6} - \frac{a_1}{a_6} f(\lambda_P^{(j)}) \right) \\ -\lambda_P^{(j)} f(\lambda_P^{(j)}) \\ 1 \end{bmatrix},$$

$$\mathbf{E}_{u,SV}^{(j)} = \begin{bmatrix} \frac{a_1}{a_6} - \frac{a_1^2+a_2a_6}{a_6} f(\lambda_{SV}^{(j)}) \\ \lambda_{SV}^{(j)} \left(\frac{1}{a_6} - \frac{a_1}{a_6} f(\lambda_{SV}^{(j)}) \right) \\ -\lambda_{SV}^{(j)} f(\lambda_{SV}^{(j)}) \\ 1 \end{bmatrix}, \tag{C12}$$

where $\mathbf{E}_{d,P}^{(j)}$, $\mathbf{E}_{d,SV}^{(j)}$, $\mathbf{E}_{u,P}^{(j)}$ and $\mathbf{E}_{u,SV}^{(j)}$ are four eigenvectors of the matrix $\mathbf{A}^{(j)}$, and correspond to the eigenvalues $-\lambda_P^{(j)}$, $-\lambda_{SV}^{(j)}$, $\lambda_P^{(j)}$ and $\lambda_{SV}^{(j)}$, respectively; the even function $f(x)$ can be defined as:

$$f(x) = \frac{a_1a_3 + a_4a_6 - x^2}{a_1^2a_3 + a_2a_3a_6 - a_1x^2},$$

$$\text{or} \quad f(x) = \frac{a_1a_5 - a_3a_6}{a_1^2a_5 + a_2a_5a_6 - a_6x^2},$$

$$\text{or} \quad f(x) = \frac{a_1a_3 + a_1a_5 + a_4a_6 - a_3a_6 - x^2}{a_1^2a_3 + a_1^2a_5 + a_2a_3a_6 + a_2a_5a_6 - (a_1 + a_6)x^2}. \tag{C13}$$

Similarly, for the SH wave case, the matrix $\mathbf{A}^{(j)}$ can be written as:

$$\mathbf{A}^{(j)} = \begin{bmatrix} 0 & a_1 \\ a_2 & 0 \end{bmatrix} \tag{C14}$$

with

$$a_1 = \frac{1}{M_{55}^{(j)}}, \; a_2 = M_{66}^{(j)}k^2 - \rho^{(j)}\omega^2. \tag{C15}$$

$\mathbf{E}^{(j)}$ and $\mathbf{\Lambda}^{(j)}$ are two matrices related to the eigendecomposition of matrix $\mathbf{A}^{(j)}$. Matrix $\mathbf{\Lambda}^{(j)}$ has the form:

$$\Lambda^{(j)}(z) = \begin{bmatrix} \Lambda_d^{(j)}(z) & 0 \\ 0 & \Lambda_u^{(j)}(z) \end{bmatrix}, \tag{C16}$$

$$\Lambda_d^{(j)}(z) = \exp\left[-\lambda_{SH}^{(j)} \left(z - z^{(j-1)} \right) \right], \tag{C17}$$

$$\Lambda_u^{(j)}(z) = \exp\left[-\lambda_{SH}^{(j)} \left(z^{(j)} - z \right) \right], \tag{C18}$$

where $-\lambda_{SH}^{(j)}$ and $\lambda_{SH}^{(j)}$ are two eigenvalues of matrix $\mathbf{A}^{(j)}$, and $\lambda_{SH}^{(j)}$ can be calculated by:

$$\lambda_{SH}^{(j)} = \sqrt{a_1a_2}, \; \mathrm{Re}[\lambda_{SH}^{(j)}] > 0. \tag{C19}$$

Matrix $\mathbf{E}^{(j)}$ is composed of two eigenvectors of matrix $\mathbf{A}^{(j)}$ corresponding to the eigenvalues $-\lambda_{SH}^{(j)}$ and $\lambda_{SH}^{(j)}$, respectively. It can be expressed as:

$$\mathbf{E}^{(j)} = \begin{bmatrix} \mathbf{E}_{d,SH}^{(j)} & \mathbf{E}_{u,SH}^{(j)} \end{bmatrix} = \begin{bmatrix} g\left(-\lambda_{SH}^{(j)}\right) & g\left(\lambda_{SH}^{(j)}\right) \\ 1 & 1 \end{bmatrix}, \tag{C20}$$

where the function $g(x)$ can be defined as:

$$g(x) = \frac{a_1}{x}, \; \mathrm{org}(x) = \frac{x}{a_2}, \; \mathrm{org}(x) = \frac{x - a_1}{a_2 - x}. \tag{C21}$$

APPENDIX D: EIGENDISPLACEMENTS AND EIGENTRACTIONS

Let's first consider the *P*-SV wave case. According to eq. (27), $\mathbf{C}_d^{(1)} = [C_{Pd}^{(1)}, C_{SVd}^{(1)}]^T$ can take any non-zero value, and without any loss of generality, it can be taken unity. Thus, for each horizontal wavenumber of Rayleigh waves (k_n, $n = 0, 1, 2, \ldots$), we find (Chen 1993):

$$\begin{aligned} \mathbf{C}_d^{(1)}(k_n) &= \left[C_{Pd}^{(1)}(k_n), C_{SVd}^{(1)}(k_n) \right]^T \\ &= \frac{1}{\sqrt{|1 - F_{11}(k_n)|^2 + |F_{12}(k_n)|^2}} [F_{12}(k_n), 1 - F_{11}(k_n)]^T, \end{aligned} \tag{D1}$$

by the conditions

$$\begin{cases} [1 - F_{11}(k_n)] C_{Pd}^{(1)}(k_n) - F_{12}(k_n) C_{SVd}^{(1)}(k_n) = 0 \\ \left[C_{Pd}^{(1)}(k_n) \right]^2 + \left[C_{SVd}^{(1)}(k_n) \right]^2 = 1 \end{cases}, \tag{D2}$$

where $\mathbf{F}(k_n) = \hat{\mathbf{R}}_{ud}^{(0)}(k_n)\hat{\mathbf{R}}_{du}^{(1)}(k_n)$. Then, we obtain:

$$\mathbf{C}_u^{(1)}(k_n) = \hat{\mathbf{R}}_{du}^{(1)}(k_n)\mathbf{C}_d^{(1)}(k_n). \tag{D3}$$

Once $\mathbf{C}_d^{(1)}$ and $\mathbf{C}_u^{(1)}$ are found, the other unknown coefficients for Rayleigh waves can be determined as follows:

$$\begin{cases} \mathbf{C}_d^{(j+1)}(k_n) = \hat{\mathbf{T}}_d^{(j)}(k_n)\hat{\mathbf{T}}_d^{(j-1)}(k_n) \cdots \hat{\mathbf{T}}_d^{(1)}(k_n)\mathbf{C}_d^{(1)}(k_n) \\ \mathbf{C}_u^{(j+1)}(k_n) = \hat{\mathbf{R}}_{du}^{(j+1)}(k_n)\mathbf{C}_d^{(j+1)}(k_n) \end{cases}, \tag{D4}$$

for $j = 1, 2, \ldots, N-1$, and

$$\mathbf{C}_d^{(N+1)}(k_n) = \hat{\mathbf{T}}_d^{(N)}(k_n)\hat{\mathbf{T}}_d^{(N-1)}(k_n) \cdots \hat{\mathbf{T}}_d^{(1)}(k_n)\mathbf{C}_d^{(1)}(k_n). \tag{D5}$$

Finally, substituting the solved $\mathbf{C}_d^{(j)}$ and $\mathbf{C}_u^{(j)}$ into eq. (B12), we can obtain the eigendisplacements ($u_S^{(j)}$ and $u_R^{(j)}$) and eigentractions ($\tau_S^{(j)}$ and $\tau_R^{(j)}$) of Rayleigh waves.

For SH wave case, $C_d^{(1)}$ and $C_u^{(1)}$ can be determined by:

$$\begin{cases} C_d^{(1)}(k_n) = 1 \\ C_u^{(1)}(k_n) = \hat{R}_{du}^{(1)}(k_n) \end{cases}. \tag{D6}$$

Then, the other unknown coefficients for Love waves can be obtained by:

$$\begin{cases} C_d^{(j+1)}(k_n) = \hat{T}_d^{(j)}(k_n)\hat{T}_d^{(j-1)}(k_n) \cdots \hat{T}_d^{(1)}(k_n)C_d^{(1)}(k_n) \\ C_u^{(j+1)}(k_n) = \hat{R}_{du}^{(j+1)}(k_n)C_d^{(j+1)}(k_n) \end{cases}, \tag{D7}$$

for $j = 1, 2, \ldots, N-1$, and

$$C_d^{(N+1)}(k_n) = \hat{T}_d^{(N)}(k_n)\hat{T}_d^{(N-1)}(k_n) \cdots \hat{T}_d^{(1)}(k_n)C_d^{(1)}(k_n). \tag{D8}$$

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Finally, substituting the solved $C_d^{(j)}$ and $C_u^{(j)}$ into eq. (B12), we can obtain the eigendisplacement ($u_T^{(j)}$) and eigentraction ($\tau_T^{(j)}$) of Love waves.

APPENDIX E: ELLIPTICITY OF RAYLEIGH WAVES

The theoretical HVSR curves (i.e. ellipticity) of Rayleigh waves can be calculated by solving the eigenproblem of Rayleigh waves in the laterally homogeneous layered media. After determining the horizontal and vertical eigendisplacements (i.e. u_S and u_R) of Rayleigh waves (see Appendix D), the theoretical HVSR curves of Rayleigh waves can be calculated by (Aki & Richards 1980):

$$HVSR(f, k_n, z = 0) = \frac{\left|u_S(f, k_n, z = 0)\right|}{\left|u_R(f, k_n, z = 0)\right|}, \tag{E1}$$

where f represents frequency; k_n represents the horizontal wavenumber of the nth-mode Rayleigh waves; the depth $z = 0$ means the free surface; the operator $|\cdot|$ means taking the modulus of a complex variable or the absolute value of a real variable.

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